

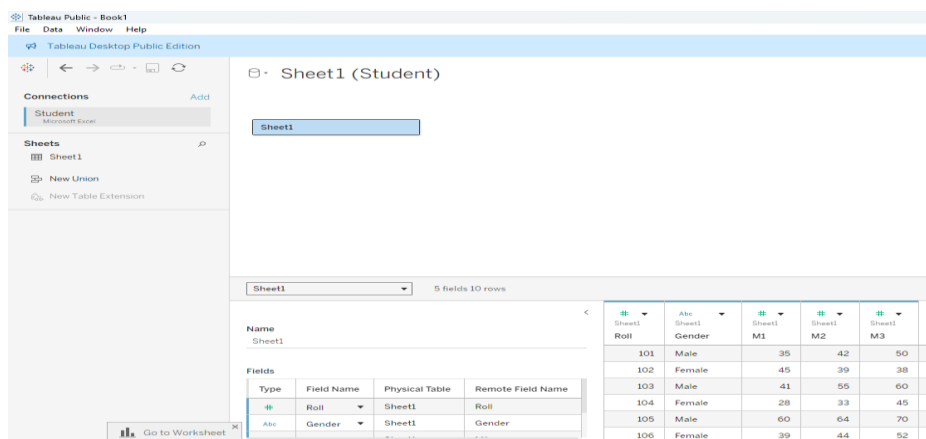
1. Create student database in excel with following data Roll no, Gender, Mark1,Mark2, Mark3 and perform the following.

Aim: importing data into tabluue and performing operations

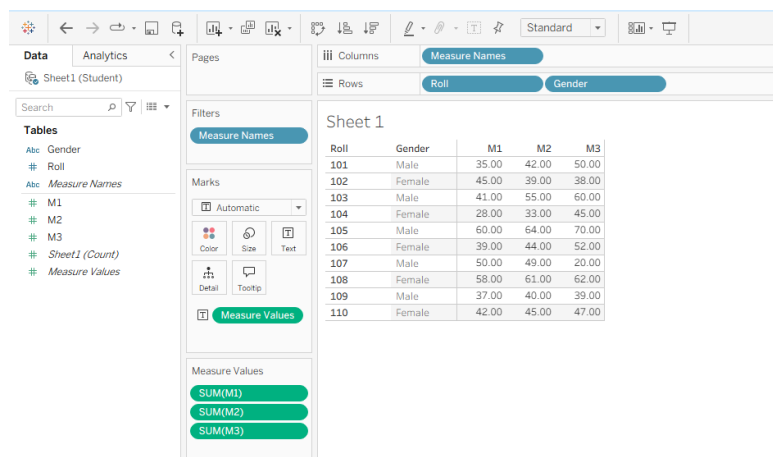
Create student database in Excel with following data Roll No Gender M1,M2,M3.

	A	B	C	D	E	F
1	Roll	Gender	M1	M2	M3	
2	101	Male	35	42	50	
3	102	Female	45	39	38	
4	103	Male	41	55	60	
5	104	Female	28	33	45	
6	105	Male	60	64	70	
7	106	Female	39	44	52	
8	107	Male	50	49	20	
9	108	Female	58	61	62	
10	109	Male	37	40	39	
11	110	Female	42	45	47	

Connect Student Excel file to Tabluue.



Display student data in a Tabluue worksheet .



a) Find the total marks of each student.

Data | Analytics <

Sheet1 (Student)

Search

- Create Calculated Field...
- Create Parameter...
- Group by Folder
- Group by Data Source Table
- Sort by Name
- Sort by Data Source Order
- Hide All Unused Fields
- Show Hidden Fields
- Expand All
- Collapse All

Total

$[M1] + [M2] + [M3]$

The calculation is valid.

Apply OK

Data | Analytics <

Sheet1 (Student)

Search

Tables

- Gender
- Roll
- Measure Names
- M1
- M2
- M3
- Total
- Sheet1 (Count)
- Measure Values

Columns | Measure Names

Rows | Roll | Gender

Marks | Measure Values

Sheet 1

Roll	Gender	M1	M2	M3	Total
101	Male	35.0	42.0	50.0	127.0
102	Female	45.0	39.0	38.0	122.0
103	Male	41.0	55.0	60.0	156.0
104	Female	28.0	33.0	45.0	106.0
105	Male	60.0	64.0	70.0	194.0
106	Female	39.0	44.0	52.0	135.0
107	Male	50.0	49.0	20.0	119.0
108	Female	58.0	61.0	62.0	181.0
109	Male	37.0	40.0	39.0	116.0
110	Female	42.0	45.0	47.0	134.0

Measure Values

- SUM(M1)
- SUM(M2)
- SUM(M3)
- SUM(Total)

b) Find the average marks of each student.

Average



[Total] / 3

The calculation is valid.

Apply

OK

Pages	Columns	Measure Names
	Rows	Roll

Filters	Sheet 1
Measure Names	
Marks	
Automatic	
Color	
Size	
Text	
Detail	
Tooltip	
Measure Va..	
Measure Values	
SUM(Average)	
SUM(M1)	
SUM(M2)	
SUM(M3)	

Roll	Average	M1	M2	M3
101	42.33	35.00	42.00	50.00
102	40.67	45.00	39.00	38.00
103	52.00	41.00	55.00	60.00
104	35.33	28.00	33.00	45.00
105	64.67	60.00	64.00	70.00
106	45.00	39.00	44.00	52.00
107	39.67	50.00	49.00	20.00
108	60.33	58.00	61.00	62.00
109	38.67	37.00	40.00	39.00
110	44.67	42.00	45.00	47.00

c) Find the result of each student with the following conditions if average > 40 then pass or fail.

Result



IF [Average] > 40 THEN "Pass" ELSE "Fail" END

The calculation is valid.

Apply

OK

# Sheet1 Roll	Abc Sheet1 Gender	# Sheet1 M1	# Sheet1 M2	# Sheet1 M3	=# Calculation Total	=# Calculation Average	=Abc Calculation Result
101	Male	35	42	50	127	42.3333	Pass
102	Female	45	39	38	122	40.6667	Pass
103	Male	41	55	60	156	52.0000	Pass
104	Female	28	33	45	106	35.3333	Fail
105	Male	60	64	70	194	64.6667	Pass
106	Female	39	44	52	135	45.0000	Pass

d) Arrange all the student's data in descending order of the total marks.

Pages

Filters

Marks

Measure Values

Columns

Rows

Measure Names

Roll
Result

Sheet 1

Roll	Result	Average	M1	M2	M3	Total
105	Pass	64.7	60.0	64.0	70.0	194.0
108	Pass	60.3	58.0	61.0	62.0	181.0
103	Pass	52.0	41.0	55.0	60.0	156.0
106	Pass	45.0	39.0	44.0	52.0	135.0
110	Pass	44.7	42.0	45.0	47.0	134.0
101	Pass	42.3	35.0	42.0	50.0	127.0
102	Pass	40.7	45.0	39.0	38.0	122.0
107	Fail	39.7	50.0	49.0	20.0	119.0
109	Fail	38.7	37.0	40.0	39.0	116.0
104	Fail	35.3	28.0	33.0	45.0	106.0

e) Find and extract student's data with minimum total marks.

Filter [Roll] ×

General Wildcard Condition Top

☐ None

☒ By field:

Bottom

1

by

Total

Minimum

☐ By formula:

Top

10

by

Reset

OK

Cancel

Apply

Pages

Columns

Measure Names

Rows

Roll

Result

Filters

Measure Names

Roll

Sheet 1

Roll	Result	Average	M1	M2	M3	Total
104	Fail	35.33	28.00	33.00	45.00	106.00

f) Find and extract student's data with maximum total marks.

Filter [Roll]

General
Wildcard
Condition
Top

☐ None
☒ By field:

Top
1
by
Total
Maximum

☐ By formula:

Top
10
by

Reset
OK
Cancel
Apply

Pages

Columns

Measure Names

Rows

Roll

Result

Filters

Measure Names

Roll

Sheet 1

Roll	Result	Average	M1	M2	M3	Total
105	Pass	64.7	60.0	64.0	70.0	194.0

g) Display only the student data with total marks greater than 150.

Filter [Roll]

General

Wildcard

Condition

Top

☐ None

☒ By field:

Total

Sum

>

150

Range of Values

Min:

Load

Max:

☐ By formula:

Reset

OK

Cancel

Apply

Pages

Columns

Measure Names

Rows

Roll

Result

Filters

Measure Names

Roll

Marks

Sheet 3

Roll	Result	Average	M1	M2	M3	Total
103	Pass	52.0	41.0	55.0	60.0	156.0
105	Pass	64.7	60.0	64.0	70.0	194.0
108	Pass	60.3	58.0	61.0	62.0	181.0

2. Create a dataset with the fields Roll No, Internal 1 (50), Internal 2 (50), Test 1 (20), Test 2 (20), Assignment 1 (20), Assignment 2 (20), and Attendance (%), and compute the internal marks of students (out of 50) by applying the given conditions using calculated fields.

a) Internal 1, Internal 2(50 marks each)- reduce to 25 marks.

b) Test 1, Test 2(20 marks each)- reduce to 10 marks.

c) Assignment 1, Assignment 2(20 marks each)-reduce to 12 marks, attendance -max of 3 marks based on condition.

a) Attendance	b) Marks
b) $\geq 92\%$	c) 3
d) $\geq 82\%$ and $< 92\%$	e) 2
f) $\geq 76\%$ and $< 82\%$	g) 1
h) $\geq 75\%$ and $< 76\%$	i) 0

d) Find the total of Internals(25), class test(10), assignment(12) and attendance marks(3) reduce this out of 50 to 30 marks.

e) Visually represent it using Horizontal bar.

3. Create a dataset using the structure provided below and import it into Tableau. Then, perform appropriate join operations to visualize and analyze the data.

- Student: RollNo, Name, Address, Phone, Age
- Student Course: course_ID, Roll_no

a) Display names and age of students enrolled in different course.

Student 1+ (student course)

Student 1 is made of 2 tables. ①

Student 1 studentcourse

Join

Inner Left Right Full Outer

Data Source studentcourse

Roll = Rollno

Add new join clause

Student 1 7 fields 5 rows 5

Name

Student 1

Fields

Type	Field Name	Physical Table	Remote Field Name
#	Roll	Student 1	ROLL
Abc	Name	Student 1	NAME

#	Abc	Abc	Abc	#	#	#
Student 1	Student 1	Student 1	Student 1	Student 1	studentcourse	studentcourse
Roll	Name	Address	Phone	AGE	Course id	Rollno
1	Harsh	Delhi	XXXXXXXXXX	18	1	1
2	Prathik	Bihar	XXXXXXXXXX	19	2	2
3	Priyanka	Siliguri	XXXXXXXXXX	20	2	3
4	Deepak	Ramnagar	XXXXXXXXXX	18	3	4
5	Saptarhi	Kolkata	XXXXXXXXXX	19	1	5

Filter [Measure Names]

General

Enter search text

☒ AGE

☐ Count of Student 1

☐ Roll

☐ Rollno

All None

Summary

Field: [Measure Names]

Student 1 is made of 2 tables. ⓘ

Student 1

student course: Student 1

studentcourse

Join

Inner

Left

Right

Full Outer

Data Source		studentcourse	
Roll	=	Rollno	
Add new join clause			

b)Display names of all students with their enrollment of course.

c)Display names of the students who enrolled for course.

d)Display all students and course enrollment details(name,course_id) from both the files.

Tableau Desktop Public Edition

Worksheet: Sheet 1

Columns: Measure Names, AGE

Rows: Course

Filters: Measure Names, AGE

Marks: Automatic

Filter (Measure Names)

General

Enter search text

☒ AGE

☐ Count of Student 1

☐ RollNo

Buy Tableau

Show Me

Let's user

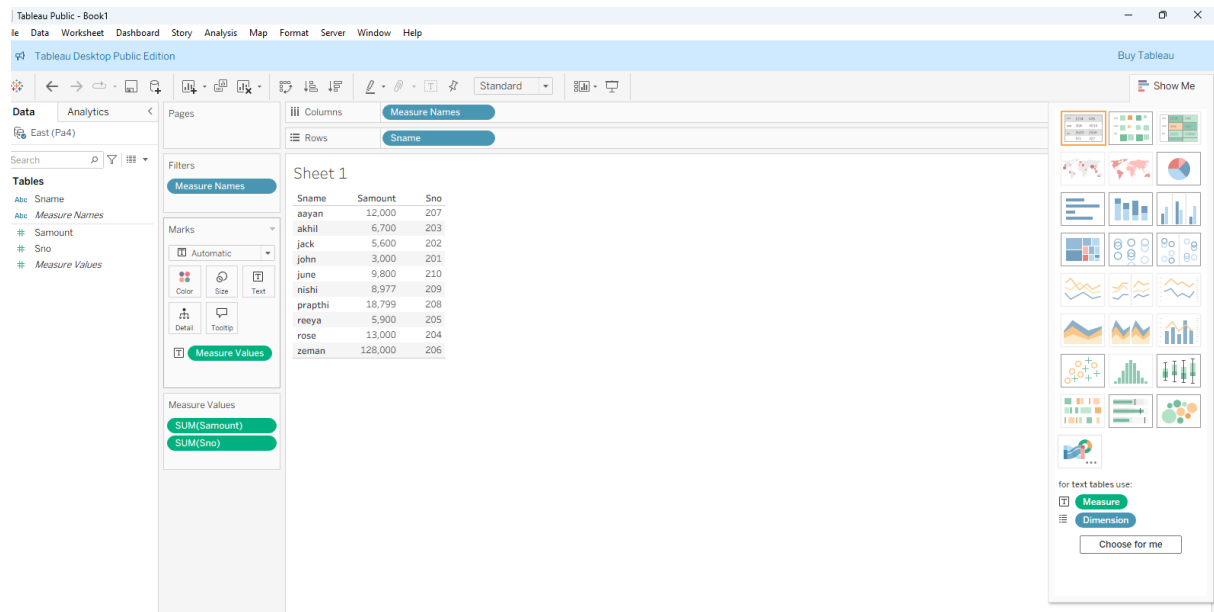
Help

Feedback

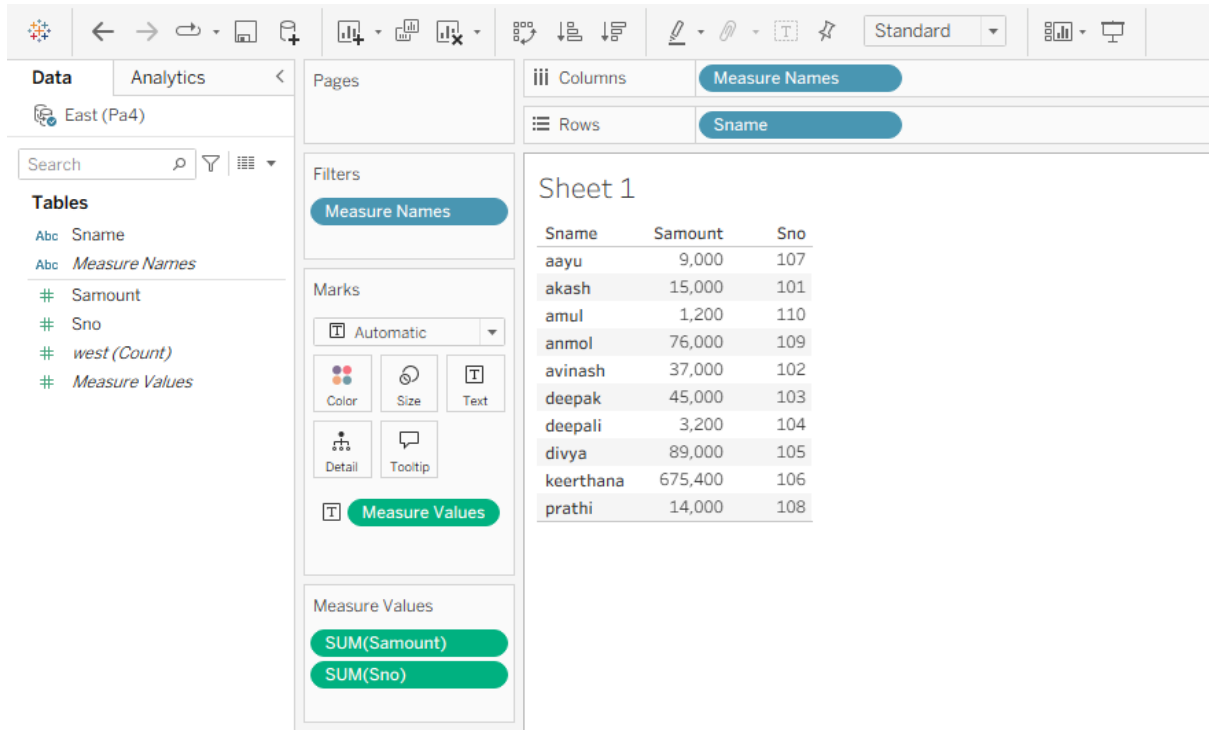
Choose for me

Sheet 1

Course id	Name	AGE	Student..	Roll	RollNo
1	Harsh	18.00	1.00	1.00	1.00
	Saptarhi	19.00	1.00	5.00	5.00
2	Prathik	19.00	1.00	2.00	2.00
	Priyanka	20.00	1.00	3.00	3.00
3	Deepak	18.00	1.00	4.00	4.00



4).



Union

✕



Specific (manual)



Wildcard (automatic)

Connection: Pa4

Drag tables here

Tables in union: 0

Apply

OK

Union

Specific (manual)

xxx

Wildcard (automatic)

Connection: Pa4

East

west

Tables in union: 2

Apply

OK

Tableau Public - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Tableau Desktop Public Edition

Data Analytics

East (Pa4)

Search

Tables

- Sheet
- Sname
- Table Name
- Measure Names
- Samount
- Sno
- Union (Count)
- Measure Values

Filters

Measure Names

Marks

Automatic

Color Size Text

Detail Tooltip

Measure Values

SUM(Samount)

SUM(Sno)

Columns

Measure Names

Rows

Sname Table Name

Sheet 1

Sname	Table Name	Samount	Sno
aayan	East	12,000	207
aayu	west	9,000	107
akash	west	15,000	101
akhil	East	6,700	203
amul	west	1,200	110
anmol	west	76,000	109
avinash	west	37,000	102
deepak	west	45,000	103
deepali	west	3,200	104
divya	west	89,000	105
jack	East	5,600	202
john	East	3,000	201
june	East	9,800	210
keerthana	west	675,400	106
nishi	East	8,977	209
prapthi	East	18,799	208
prathi	west	14,000	108
reeya	East	5,900	205
rose	East	13,000	204
zeman	East	128,000	206

Tableau Desktop Public Edition

Search

Tables

- Sheet
- Sname
- Table Name
- Measure Names
- Samount
- Sno
- Union (Count)
- Measure Values

Filters

Measure Names

Marks

Automatic

Color Size Text

Detail Tooltip

Measure Values

SUM(Samount)

SUM(Sno)

Columns

Measure Names

Rows

Sname Table Name

Sheet 1

Sname	Table Name	Sam..	Sno
keerthana	west	675,400	106
zeman	East	128,000	206
divya	west	89,000	105
anmol	west	76,000	109
deepak	west	45,000	103
avinash	west	37,000	102
prapthi	East	18,799	208
akash	west	15,000	101
prathi	west	14,000	108
rose	East	13,000	204
aayan	East	12,000	207
june	East	9,800	210
aayu	west	9,000	107
nishi	East	8,977	209
akhil	East	6,700	203
reeya	East	5,900	205
jack	East	5,600	202
deepali	west	3,200	104
john	East	3,000	201
amul	west	1,200	110

Filter [Measure Names]

General

Enter search text

☐ AGE

☐ Count of Student 1

☒ Roll

☐ Rollno

All

None

Summary

Field: [Measure Names]

Selection: Selected 1 of 4 values

Wildcard: All

Condition: None

Limit: None

Reset

OK

Cancel

Apply

Sheet 1

Course id	Name	Roll
Null	Dhanraj	6.000
	Niraj	8.000
	Rohith	7.000
1	Harsh	1.000
	Saptarhi	5.000
2	Prathik	2.000
	Priyanka	3.000
3	Deepak	4.000

Student 1+ (student course)

Filters
0 | Add

Student 1 is made of 2 tables. ⓪

×

Student 1 studentcourse

Join

Inner Left Right Full Outer

Data Source		studentcourse
Roll	=	Rollno
Add new join clause		

Sheet 1

Course id	Name	Roll
1	Harsh	1.000
	Saptarhi	5.000
2	Prathik	2.000
	Priyanka	3.000
3	Deepak	4.000
4	Null	
5	Null	

Student 1+ (student course)

Student 1 is made of 2 tables. ⓘ

Student 1 studentcourse

Join

 Inner

 Left

 Right

 Full Outer

Data Source		studentcourse
Roll	=	Rollno
Add new join clause		

Filter [Measure Names] X

General

Enter search text

☒ AGE
☐ Count of Student 1
☒ Roll
☐ Rollno

All None

Summary

Field: [Measure Names]
Selection: Selected 2 of 4 values
Wildcard: All
Condition: None
Limit: None

Reset OK Cancel Apply

Columns	Measure Names			
Rows	Course id	Name	Address	

Sheet 1

Course id	Name	Address	AGE	Roll
Null	Dhanraj	Barabjar	20.00	6.00
	Niraj	Alipur	19.00	8.00
	Rohith	Balurghat	18.00	7.00
1	Harsh	Delhi	18.00	1.00
	Saptarhi	Kolkata	19.00	5.00
2	Prathik	Bihar	19.00	2.00
	Priyanka	Siliguri	20.00	3.00
3	Deepak	Ramnagar	18.00	4.00
4	Null	Null		
5	Null	Null		

Filter [Samount]

Close

Range of values

At least

At most

Special

At least

50,000

6,75,400

1,200

6,75,400

Show: Only Relevant Values

☐ Include Null Values

Reset OK Cancel Apply

Filter [Sname] X

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

- ☒ aayan
- ☒ aayu
- ☒ akash
- ☒ akhil
- ☒ amul
- ☒ anmol
- ☒ avinash
- ☒ deepak
- ☒ deepali
- ☒ divya
- ☒ jack

All None ☐ Exclude

Summary

Field: [Sname]
Selection: Selected 20 of 20 values
Wildcard: All
Condition: None
Limit: None

Reset OK Cancel Apply

Filter [Sname] X

General Wildcard Condition Top

☐ None

☒ By field:

Top 5 by

Samount Sum

☐ By formula:

Top 10 by

Reset OK Cancel Apply

Data Analytics < Pages Columns Measure Names
 Rows Sname Table Name

East (Pa4)

Search

Tables

- Sheet
- Sname
- Table Name
- Measure Names
- Samount
- Sno
- Union (Count)
- Measure Values

Filters

- Measure Names
- Sname

Marks

Automatic

Color Size Text

Detail Tooltip

Measure Values

- SUM(Samount)
- SUM(Sno)

Sheet 1

Sname	Table Name	Sam..	Sno
keerthana	west	675,400	106
zeman	East	128,000	206
divya	west	89,000	105
anmol	west	76,000	109
deepak	west	45,000	103

5.

	A	B	C	D	E	F
1	EmpID	Name	Department	Expereince(years)	Salary	Rating
2	101	Asha	HR	3	40000	4
3	102	Raj	IT	5	60000	5
4	103	Meena	Finance	2	35000	3
5	104	Ravi	IT	6	70000	5
6	105	Neha	Marketing	1	30000	2
7	106	John	HR	4	45000	4
8	107	Anjali	Finance	7	80000	5
9	108	Samir	IT	2	50000	3
10	109	Tina	Marketing	5	55000	4
11	110	Kiran	hr	8	75000	5
12	111	Sneha	IT	1	42000	3
13	112	Manish	Finance	4	47000	4
14	113	Pooja	Marketing	6	62000	5
15	114	Arjun	HR	2	39000	3
16	115	Divya	Finance	5	68000	5
17	116	Vinay	IT	7	72000	4
18	117	Riya	Marketing	3	49000	3
19	118	Ajay	HR	6	61000	5
20	119	Kavya	Finance	1	34000	2
21	120	Varun	IT	4	53000	4

Show Start Page (Ctrl+2)

Data Analytics

Sheet1 (pa5)

Search

Tables

- Department
- Emp ID
- Name
- Measure Names
- Bonous
- Expereince(years)
- Rating
- Salary
- Sheet1 (Count)
- Measure Values

Filters

Measure Names

Marks

Automatic

Color Size Text

Detail Tooltip

Measure Values

SUM(Bonous)
 SUM(Expereince(ye...)
 SUM(Rating)
 SUM(Salary)

Columns

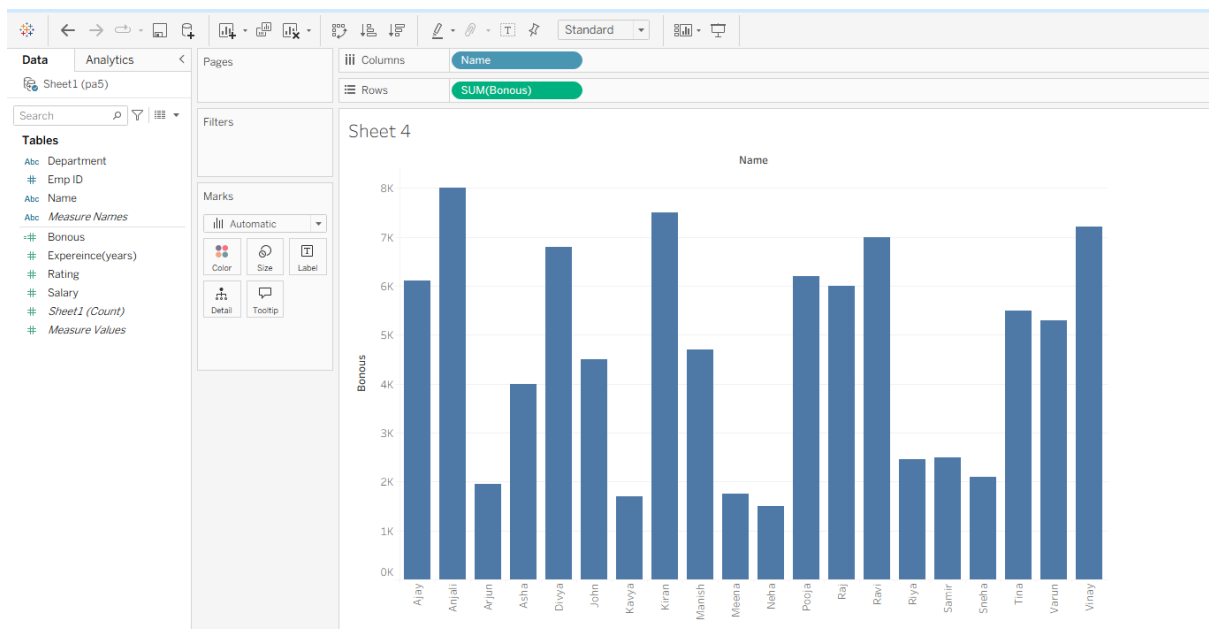
Measure Names

Rows

Department Emp ID Name

Sheet 2

Department	Emp ID	Name	Bonous	Experei..	Rating	Salary
Finance	103	Meena	1,750	2	3	35,000
	107	Anjali	8,000	7	5	80,000
	112	Manish	4,700	4	4	47,000
	115	Divya	6,800	5	5	68,000
HR	119	Kavya	1,700	1	2	34,000
	101	Asha	4,000	3	4	40,000
	106	John	4,500	4	4	45,000
	110	Kiran	7,500	8	5	75,000
IT	114	Arjun	1,950	2	3	39,000
	118	Ajay	6,100	6	5	61,000
	102	Raj	6,000	5	5	60,000
	104	Ravi	7,000	6	5	70,000
Marketing	108	Samir	2,500	2	3	50,000
	111	Sneha	2,100	1	3	42,000
	116	Vinay	7,200	7	4	72,000
	120	Varun	5,300	4	4	53,000
Marketing	105	Neha	1,500	1	2	30,000
	109	Tina	5,500	5	4	55,000
	113	Pooja	6,200	6	5	62,000
	117	Riya	2,450	3	3	49,000



Pages

Columns

Measure Names

Rows

Emp ID

Name

Department

Performance category

Filters

Drag dimensions or measures here.

Measure Names

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

SUM(Bonus)

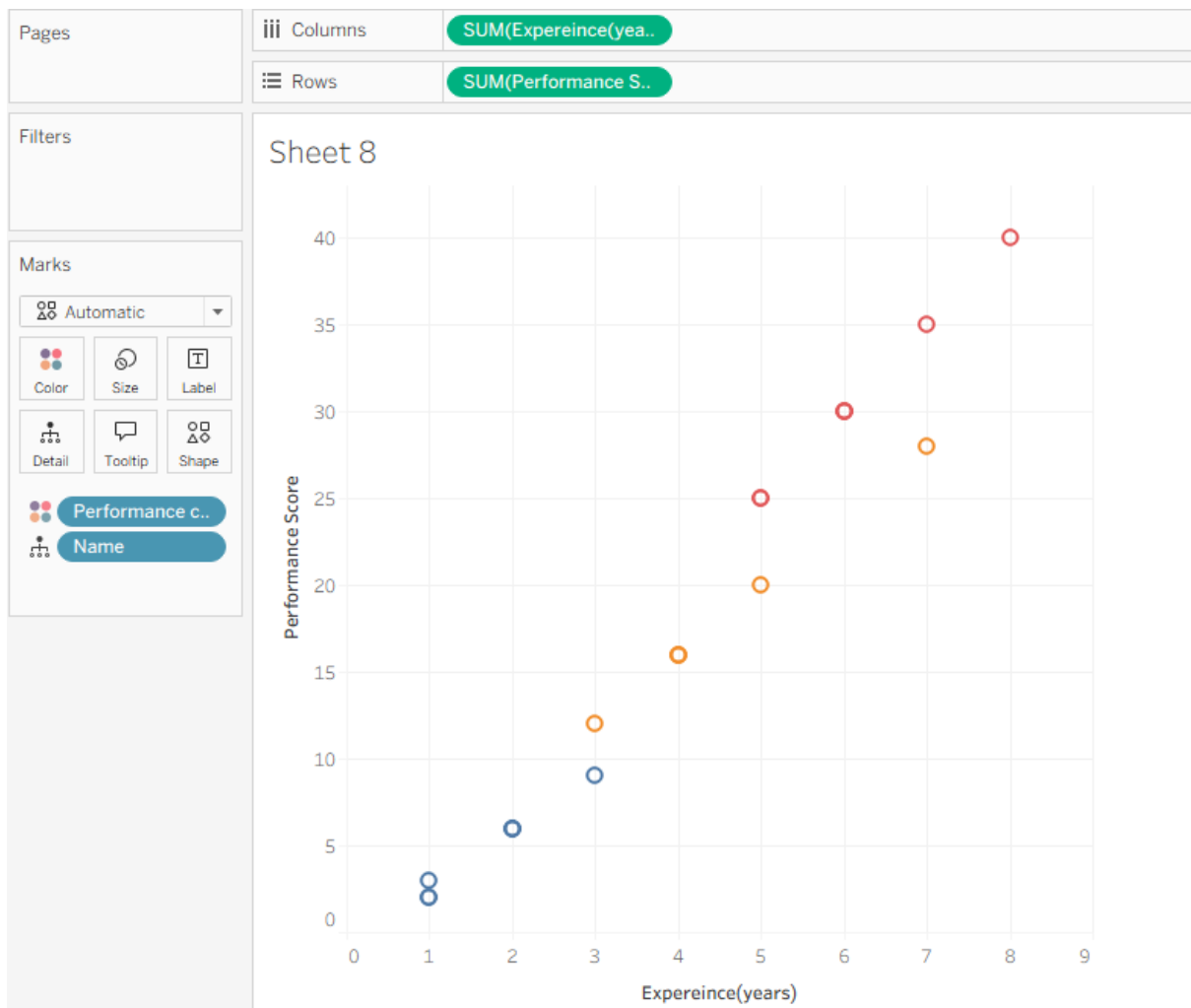
SUM(Experience(ye..

SUM(Performance S..

SUM(Rating)

SUM(Salary)

Emp ID	Name	Department	Performance category	Bonus	Experience(..	Performance Score	Rating	Salary
101	Asha	HR	Good	4,000	3	12	4	40,000
102	Raj	IT	Outstanding	6,000	5	25	5	60,000
103	Meena	Finance	Average	1,750	2	6	3	35,000
104	Ravi	IT	Outstanding	7,000	6	30	5	70,000
105	Neha	Marketing	Average	1,500	1	2	2	30,000
106	John	HR	Good	4,500	4	16	4	45,000
107	Anjali	Finance	Outstanding	8,000	7	35	5	80,000
108	Samir	IT	Average	2,500	2	6	3	50,000
109	Tina	Marketing	Good	5,500	5	20	4	55,000
110	Kiran	hr	Outstanding	7,500	8	40	5	75,000
111	Sneha	IT	Average	2,100	1	3	3	42,000
112	Manish	Finance	Good	4,700	4	16	4	47,000
113	Pooja	Marketing	Outstanding	6,200	6	30	5	62,000
114	Arjun	HR	Average	1,950	2	6	3	39,000
115	Divya	Finance	Outstanding	6,800	5	25	5	68,000
116	Vinay	IT	Good	7,200	7	28	4	72,000
117	Riya	Marketing	Average	2,450	3	9	3	49,000
118	Ajay	HR	Outstanding	6,100	6	30	5	61,000
119	Kavya	Finance	Average	1,700	1	2	2	34,000
120	Varun	IT	Good	5,300	4	16	4	53,000



6)Region

	A	B	C	D
1	Region	Target	Year	
2	South	45000	2023	
3	North	35000	2023	
4	West	25000	2024	
5	East	20000	2024	
6				
7				

Region	Product	Sales	Year
South	Laptop	50000	2023
North	Table	30000	2023
West	Phone	20000	2024
East	Laptop	25000	2024

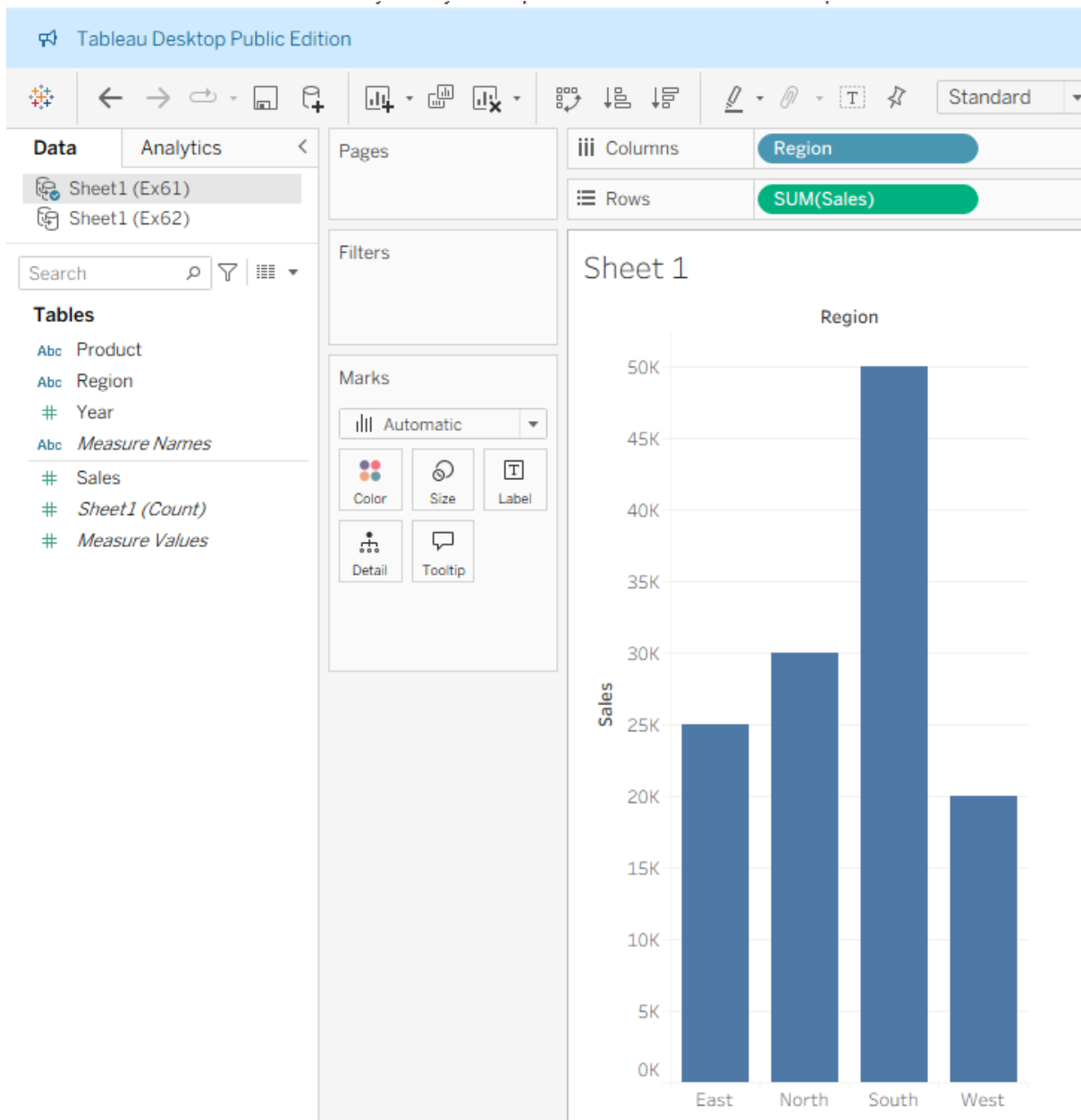
a)

The screenshot shows the Tableau Desktop Public Edition interface. The menu bar includes File, Data, Worksheet, Dashboard, Story, Analysis, Map, Format, Server, Window, and Help. The toolbar contains various icons for navigation and analysis. The data pane on the left shows a search bar and a list of tables: Product, Region, Year, Measure Names, Sales, Sheet1 (Count), and Measure Values. The marks card at the bottom shows the following configuration:

- Columns: Region
- Rows: (empty)
- Filters: (empty)
- Marks: Automatic
- Color: (empty)
- Size: (empty)
- Text: (empty)
- Detail: (empty)
- Tooltip: (empty)
- Aggregation: SUM(Sales)

The data table in the center shows the following data:

Region	Sales
East	25,000
North	30,000
South	50,000
West	20,000



b)

Power BI Desktop interface showing the Data view. The left pane displays the Data source (Sheet1 (Ex62)) and the Tables list. The main area shows the visual configuration for a table.

Tables

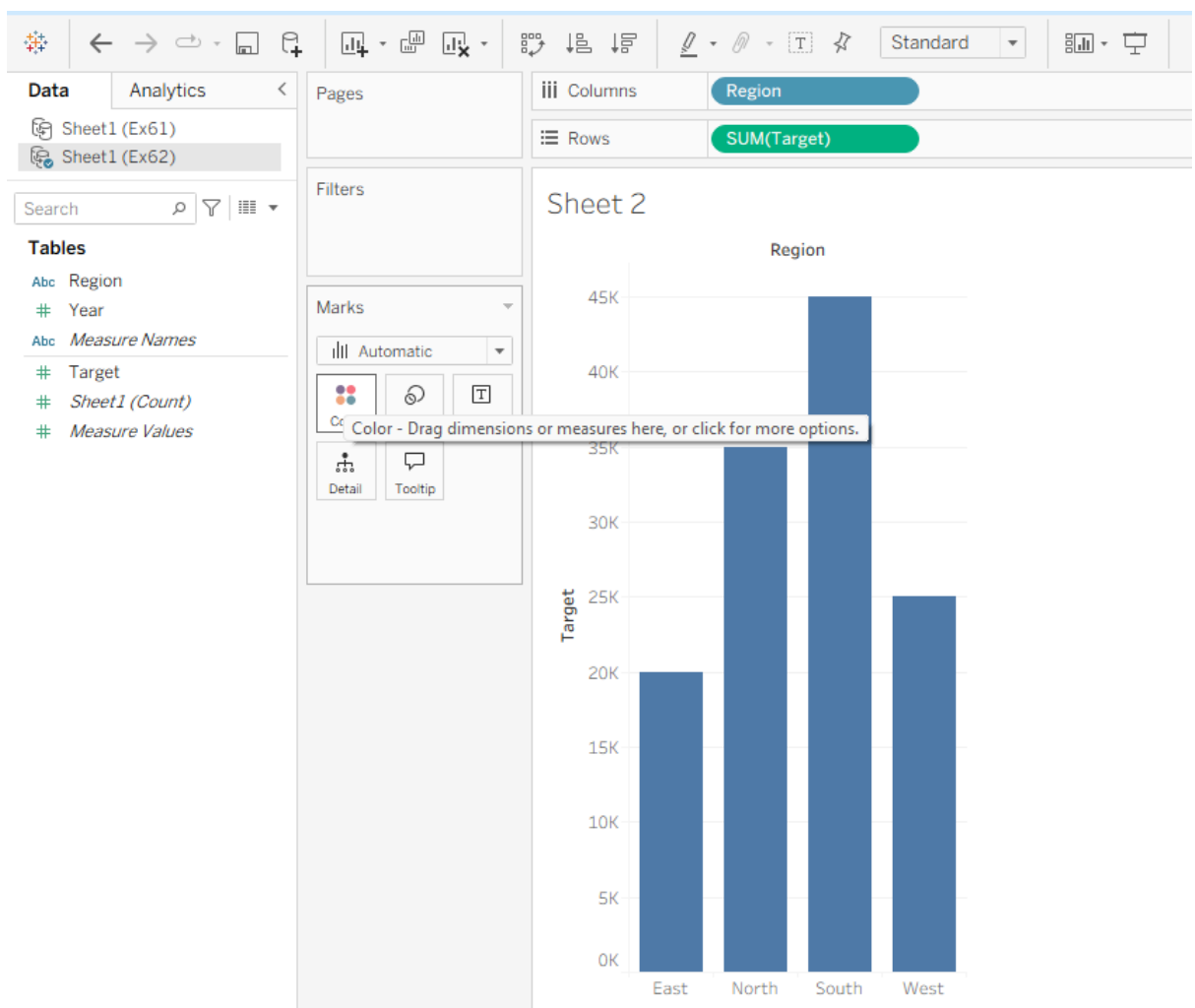
- Region
- Year
- Measure Names
- Target
- Sheet1 (Count)
- Measure Values

Visual Configuration:

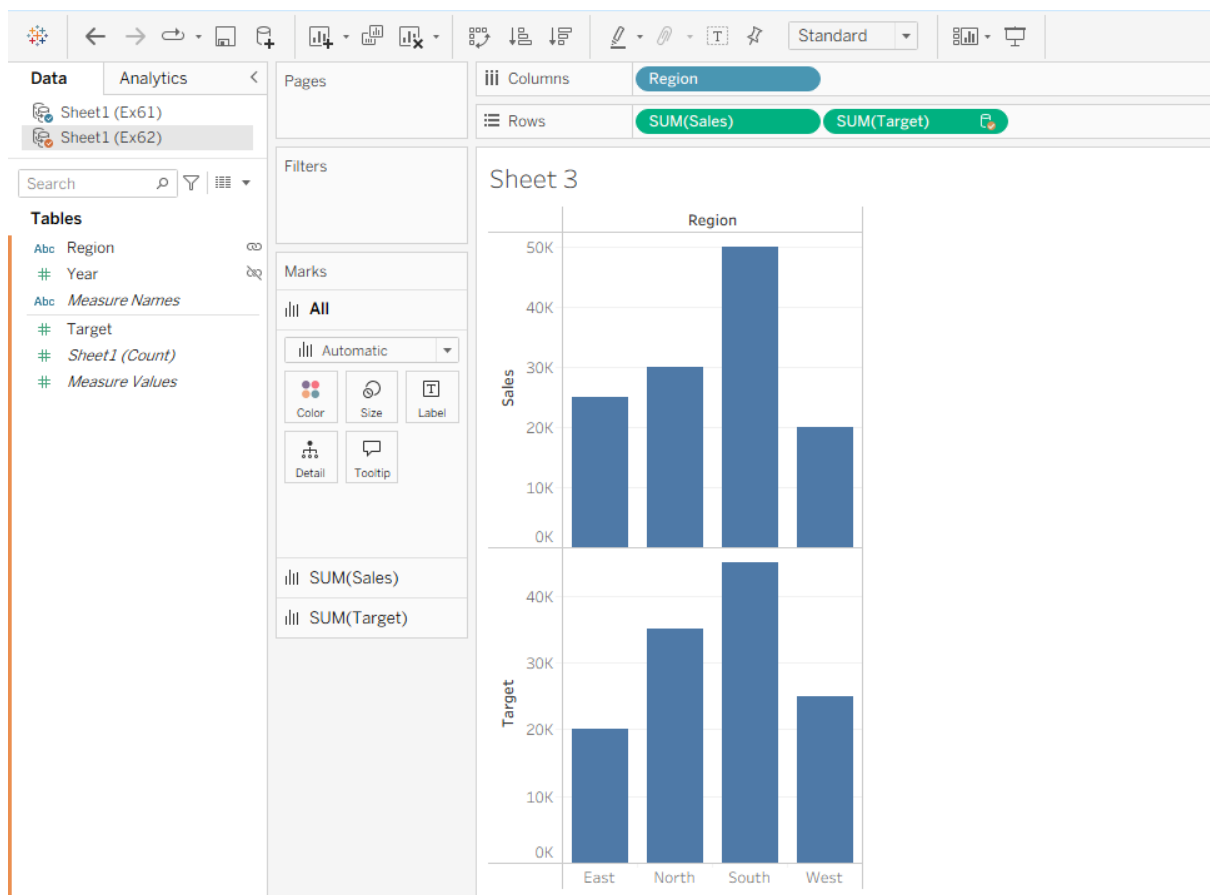
- Columns:** Region
- Rows:** (Empty)
- Filters:** (Empty)
- Marks:** SUM(Target)

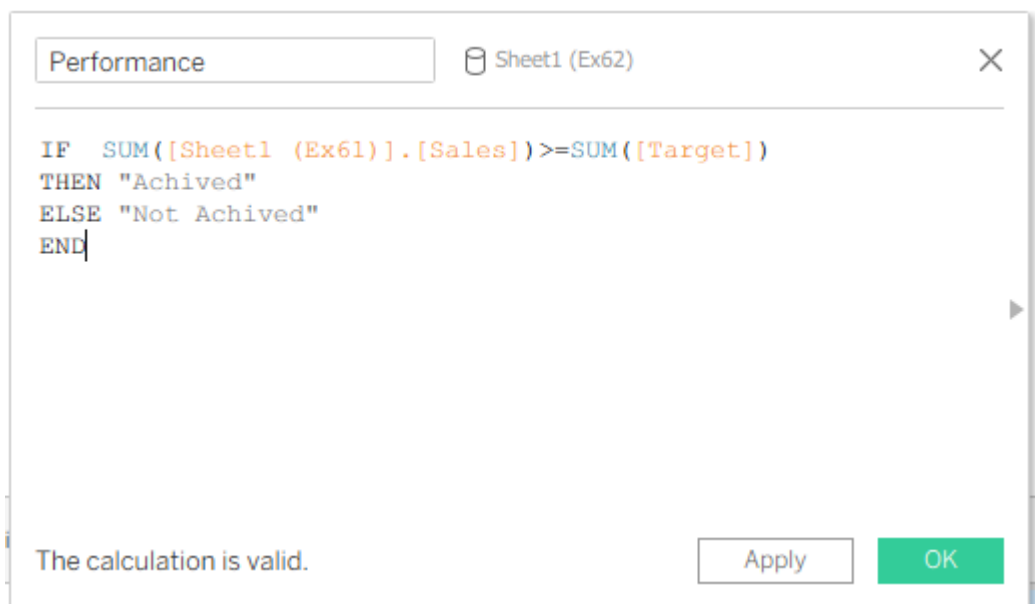
Sheet 2 Data:

Region	Value
East	20,000
North	35,000
South	45,000
West	25,000

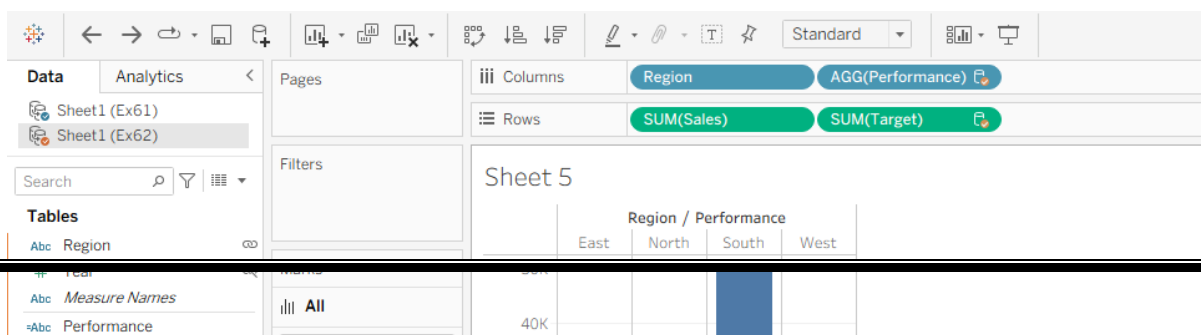


c)





d)



7.

	A	B	C	D	E
1	Product	Category	Region	Sales	Profit
2	Laptop	Electronics	North	55000	6000
3	Phone	Electronics	South	30000	4000
4	Tablet	Electronics	West	20000	1500
5	Chair	Furniture	East	15000	2000
6	Desk	Furniture	North	18000	2500
7	Monitor	Electronics	South	22000	2800
8	Sofa	Furniture	West	30000	3200
9	Mouse	Accessories	East	5000	500
10	Keyboard	Accessories	South	7000	800
11	Headphones	Accessories	North	9000	900

A)

Tableau Desktop interface showing a pivot table for Sheet 1. The interface includes a top toolbar, a left sidebar with 'Data' and 'Analytics' tabs, and a central workspace. The 'Columns' shelf contains 'Measure Names', and the 'Rows' shelf contains 'Region', 'Product', and 'Category'. The 'Marks' shelf is set to 'Automatic'. The 'Measure Values' shelf contains 'SUM(Profit)' and 'SUM(Sales)'. The pivot table shows data for four regions: East, North, South, and West, with columns for Product, Category, Profit, and Sales.

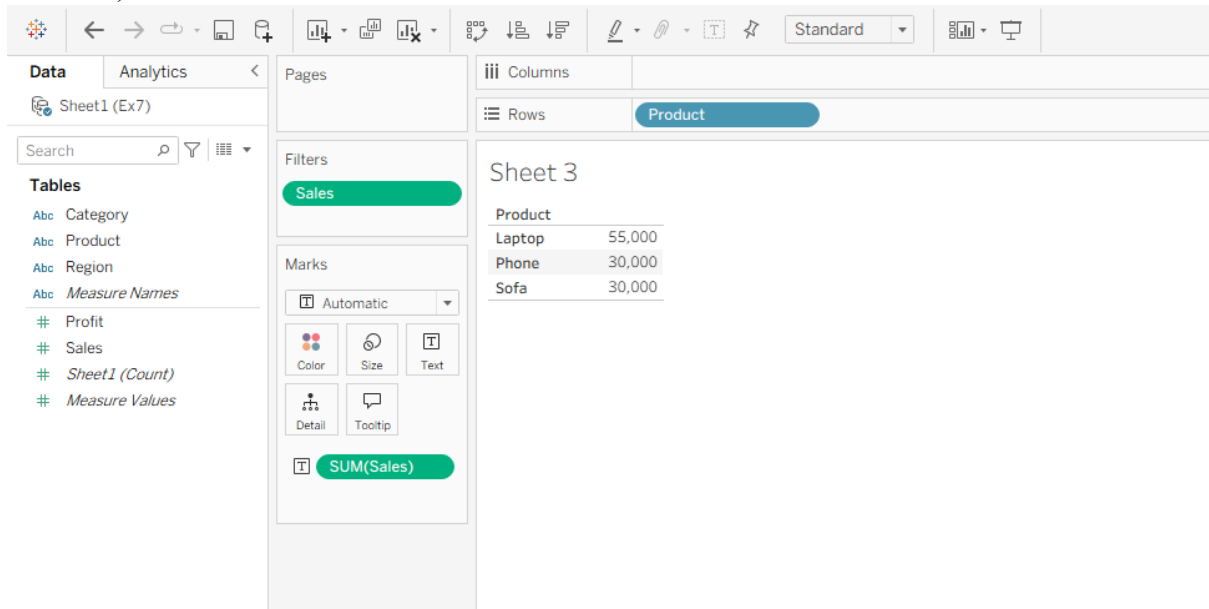
Region	Product	Category	Profit	Sales
East	Chair	Furniture	2,000	15,000
	Mouse	Accessories	500	5,000
North	Desk	Furniture	2,500	18,000
	Headphones	Accessories	900	9,000
	Laptop	Electronics	6,000	55,000
South	Keyboard	Accessories	800	7,000
	Monitor	Electronics	2,800	22,000
	Phone	Electronics	4,000	30,000
West	Sofa	Furniture	3,200	30,000
	Tablet	Electronics	1,500	20,000

b)

Tableau Desktop interface showing a pivot table for Sheet 2. The interface is similar to the previous one, but the 'Filters' shelf now contains 'Category: Electronics'. The 'Columns' shelf contains 'Measure Names', and the 'Rows' shelf contains 'Region', 'Product', and 'Category'. The 'Marks' shelf is set to 'Automatic'. The 'Measure Values' shelf contains 'SUM(Profit)' and 'SUM(Sales)'. The pivot table shows data for three regions: North, South, and West, with columns for Product, Category, Profit, and Sales.

Region	Product	Category	Profit	Sales
North	Laptop	Electronics	6,000	55,000
South	Monitor	Electronics	2,800	22,000
	Phone	Electronics	4,000	30,000
West	Tablet	Electronics	1,500	20,000

c)



d)

Filter [Product] ✕

General Wildcard Condition Top

Match value: ☐ Exclude

phone

☒ Contains Clear

☐ Starts with

☐ Ends with

☐ Exactly matches

☒ Include all values when empty

Reset OK Cancel Apply

Tableau Desktop Public Edition

Search

Tables

- Category
- Product
- Region
- Measure Names
- Profit
- Sales
- Sheet1 (Count)
- Measure Values

Columns

Measure Names

Rows

Product Category

Filters

Measure Names

Product

Marks

Automatic

Color Size Text

Detail Tooltip

Measure Values

Measure Values

CNT(Sheet1)

SUM(Profit)

SUM(Sales)

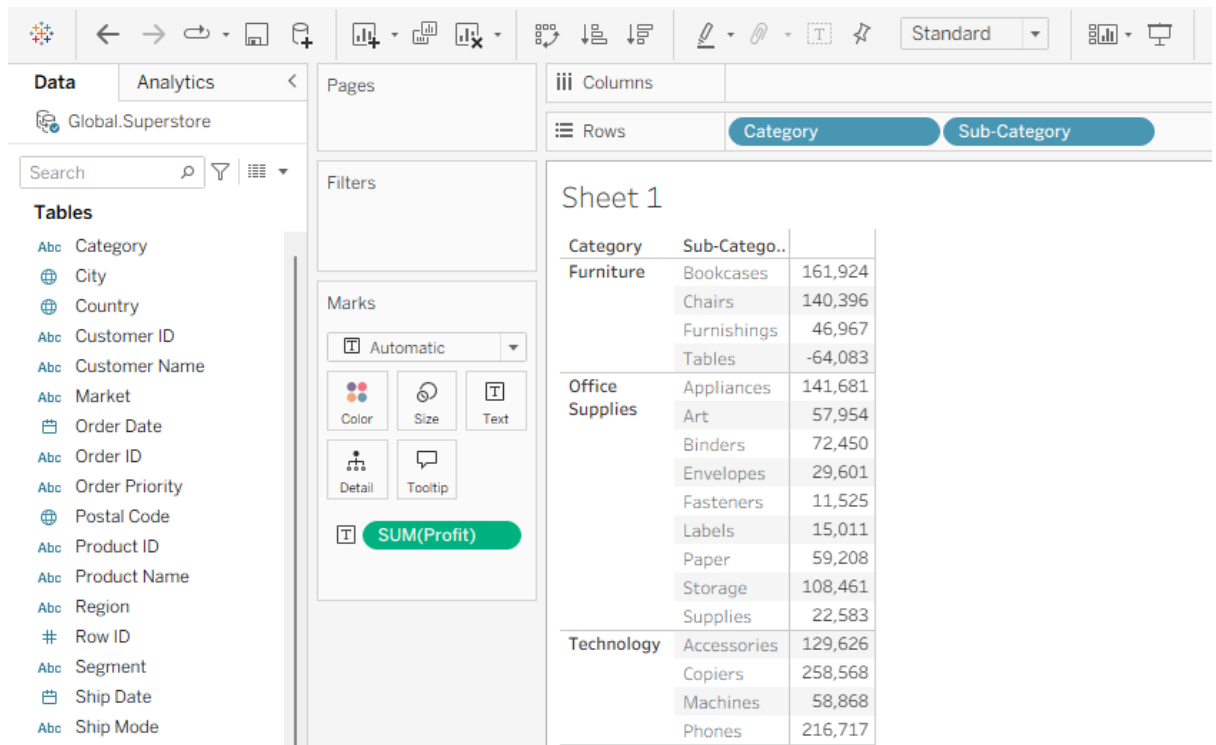
Sheet 5

Product	Category	Count of Sheet1	Profit	Sales
Headphones	Accessories	Abc	Abc	Abc
Phone	Electronics	Abc	Abc	Abc

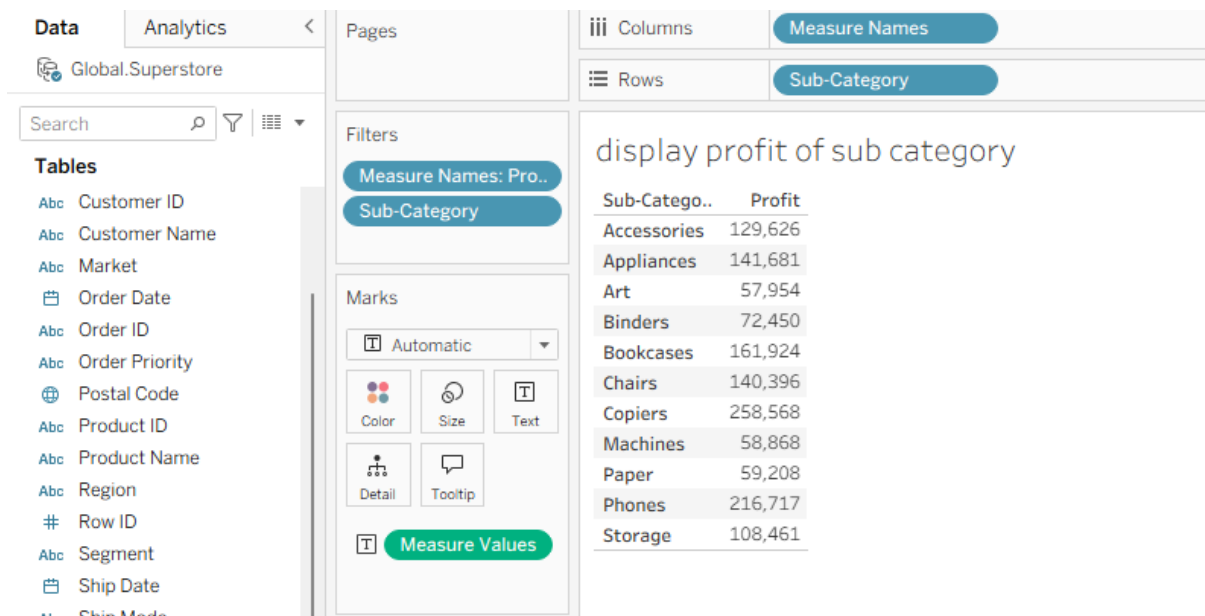
Part b

1)

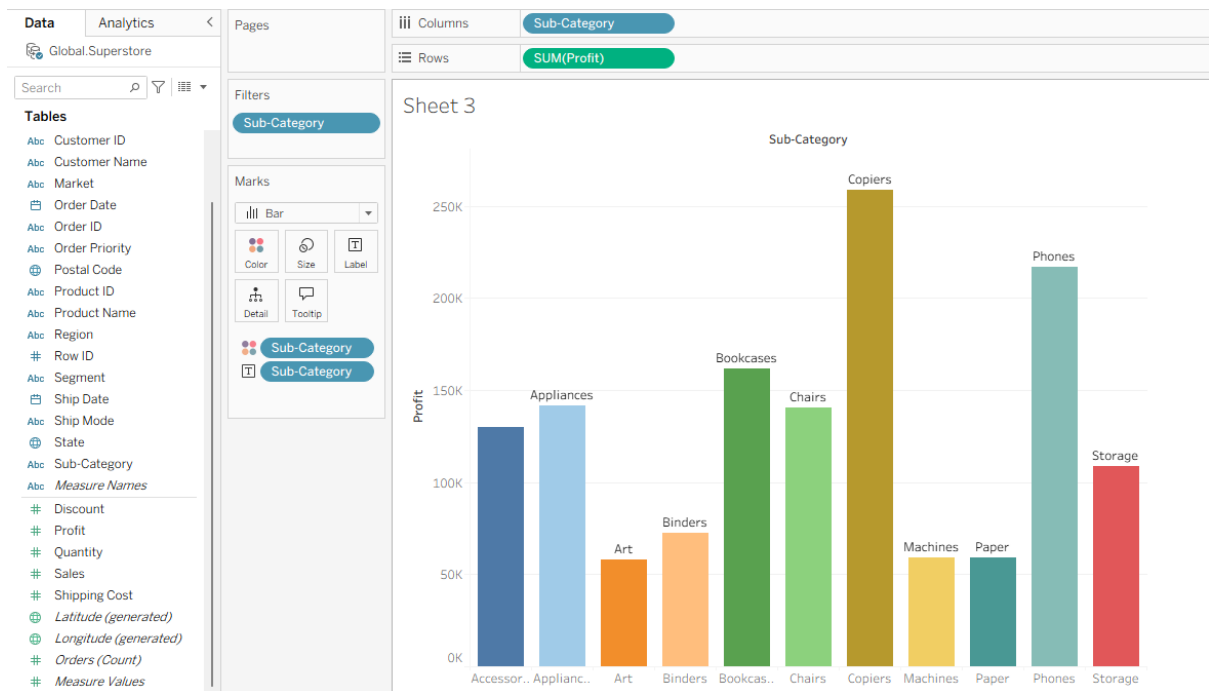
a)



b)



c)



Data | Analytics | Pages

Global.Superstore

Search

Tables

- Abc Category
- City
- Country
- Abc Customer ID
- Abc Customer Name
- Abc Market
- Order Date
- Abc Order ID
- Abc Order Priority
- Postal Code
- Abc Product ID
- Abc Product Name
- Abc Region
- # Row ID
- Abc Segment
- Ship Date

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

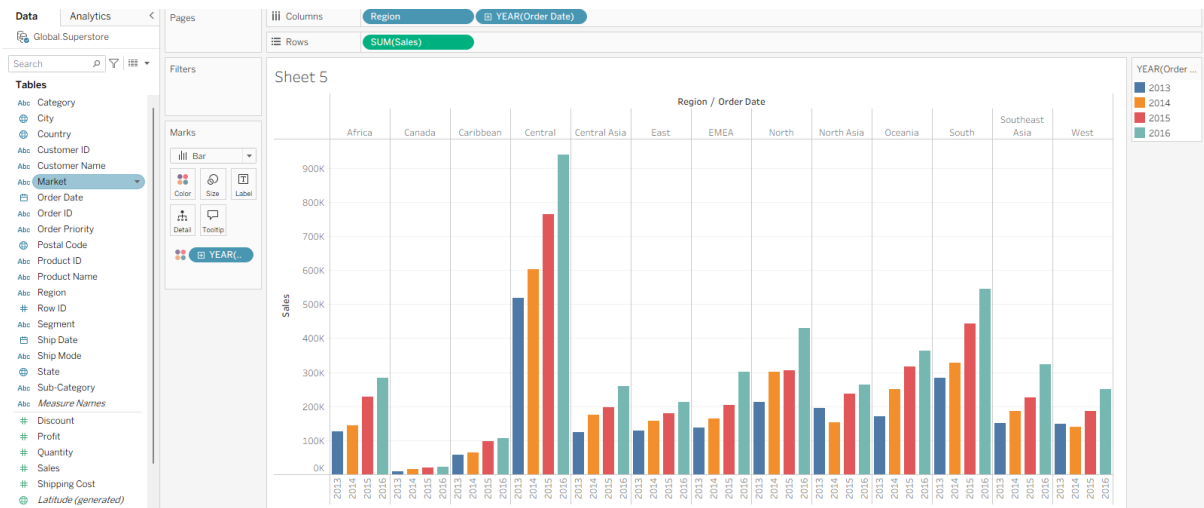
SUM(Sales)

Columns YEAR(Order Date)

Rows Region

Sheet 4

Region	Order Date			
	2013	2014	2015	2016
Africa	127,187	144,481	229,069	283,036
Canada	8,509	16,097	19,161	23,161
Caribbean	57,043	64,148	97,598	105,492
Central	517,502	602,045	763,715	939,040
Central Asia	123,238	174,133	196,315	259,140
East	128,680	156,332	180,529	213,239
EMEA	136,420	163,414	204,641	301,686
North	212,938	300,231	305,833	429,164
North Asia	194,527	152,357	236,865	264,560
Oceania	171,419	250,066	316,270	362,430
South	284,042	328,006	443,643	545,216
Southeast Asia	150,062	186,163	225,130	323,068
West	147,883	139,966	186,976	250,633



d)

saleshighlight



```
IF SUM([Sales]) = WINDOW_MAX(SUM([Sales])) THEN "Max"
ELSEIF SUM([Sales]) = WINDOW_MIN(SUM([Sales])) THEN "Min"
ELSE "other"
END
```

Default Table Calculation

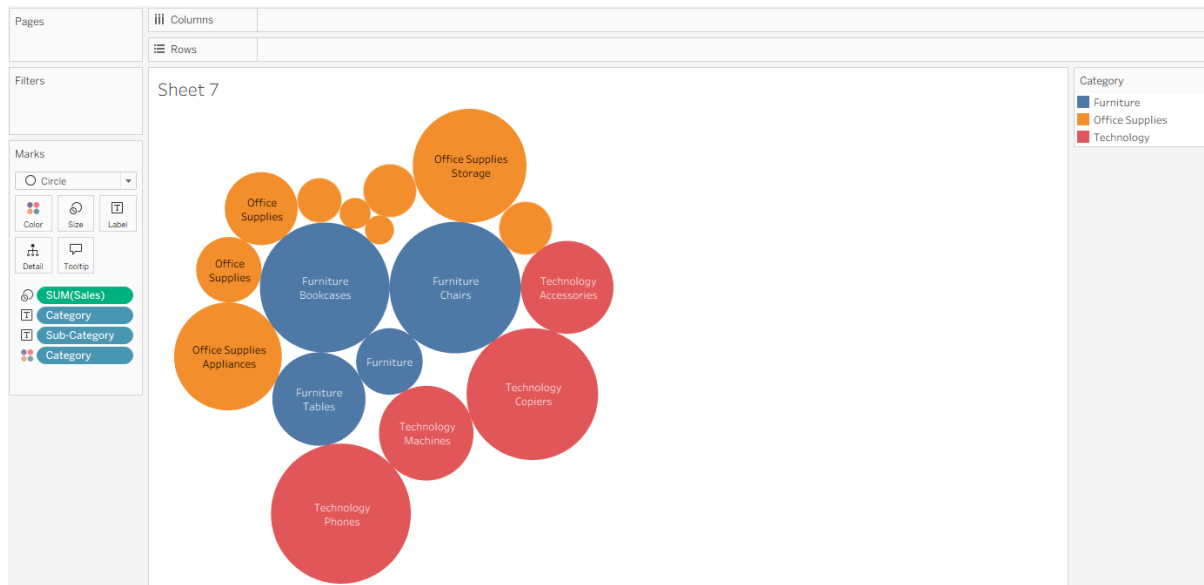
The calculation is valid.

Apply

OK

Region	2013	2014	2015	2016
Africa	127,187	144,481	229,069	283,036
Canada	8,509	16,097	19,161	23,161
Caribbean	57,043	64,148	97,598	105,492
Central	517,502	602,045	763,715	939,040
Central Asia	123,238	174,133	196,315	259,140
East	128,680	156,332	180,529	213,239
EMEA	136,420	163,414	204,641	301,686
North	212,938	300,231	305,833	429,164
North Asia	194,527	152,357	236,865	264,560
Oceania	171,419	250,066	316,270	362,430
South	284,042	328,006	443,643	545,216
Southeast Asia	150,062	186,163	225,130	323,068
West	147,883	139,966	186,976	250,633

e)



2)

×

AVG ([X])

▶

The calculation is valid.

Apply

OK

Data

Analytics

Sheet1 (anscombes-qua...)

Search

Tables

Y 1

Measure Names

corrx

Mean x

Mean y

Var x

Var y

X

X 1

X 2

X 3

Y

Y 2

Y 3

Sheet1 (Count)

Measure Values

Pages

Filters

Measure Names

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

AGG(Mean x)

AGG(Mean y)

AGG(Var x)

AGG(Var y)

AGG(corrx)

Columns

Rows

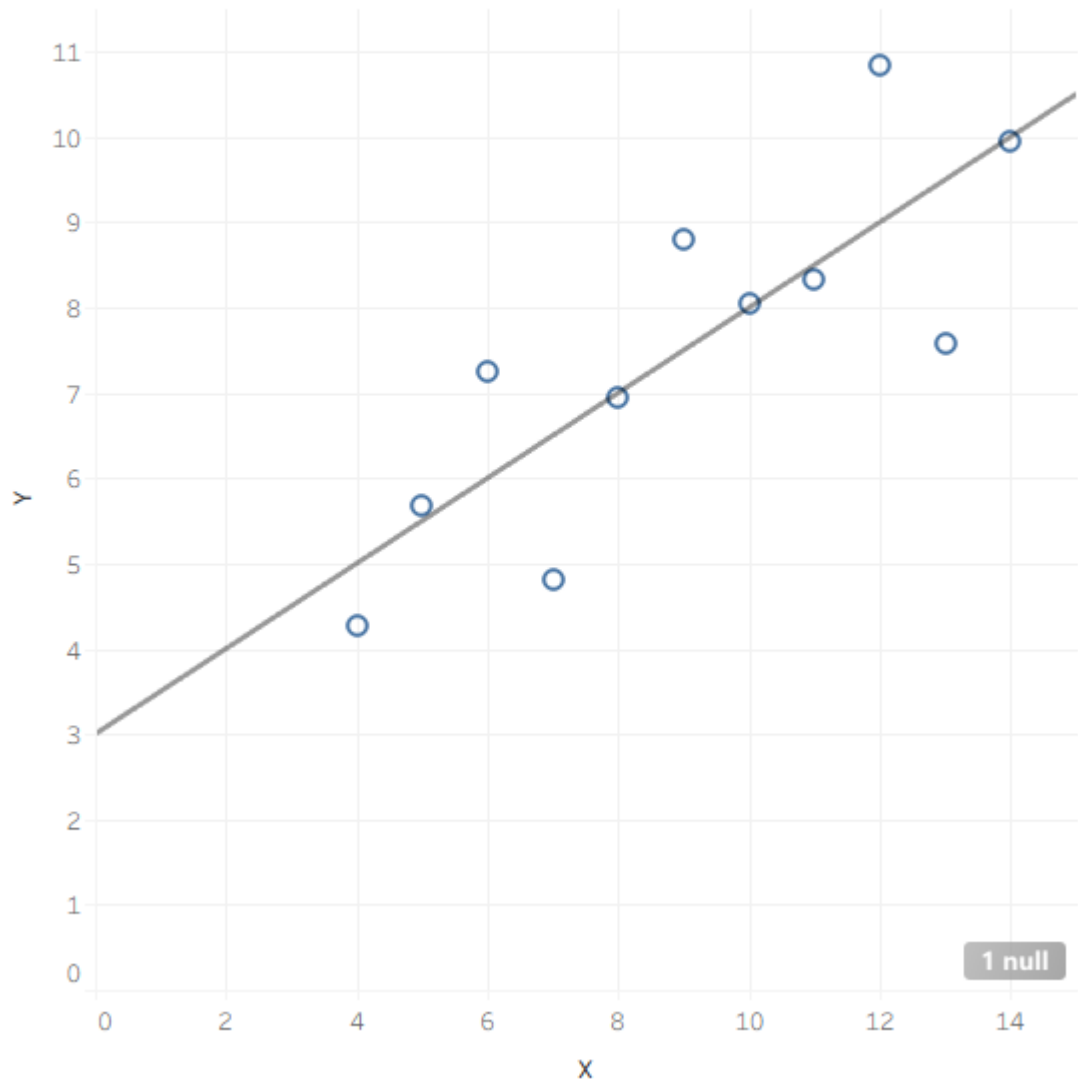
Measure Names

Sheet 1

Mean x	9.00
Mean y	7.50
Var x	11.00
Var y	4.13
corrx	0.82

Columns	X
Rows	Y

Sheet 2



Data

Analytics

Sheet1 (anscombes-qua...

Search

Tables

Abc

Measure Names

Corr X1Y1

corrxy

Mean x

Mean X1

Mean y

Mean Y1

Var x

Var X1

Var y

Var Y1

X

X 1

X 2

X 3

Y

Y 1

Y 2

Y 3

Sheet1 (Count)

Measure Values

Pages

Filters

Measure Names

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

AGG(Corr X1Y1)

AGG(Mean X1)

AGG(Mean Y1)

AGG(Var X1)

AGG(Var Y1)

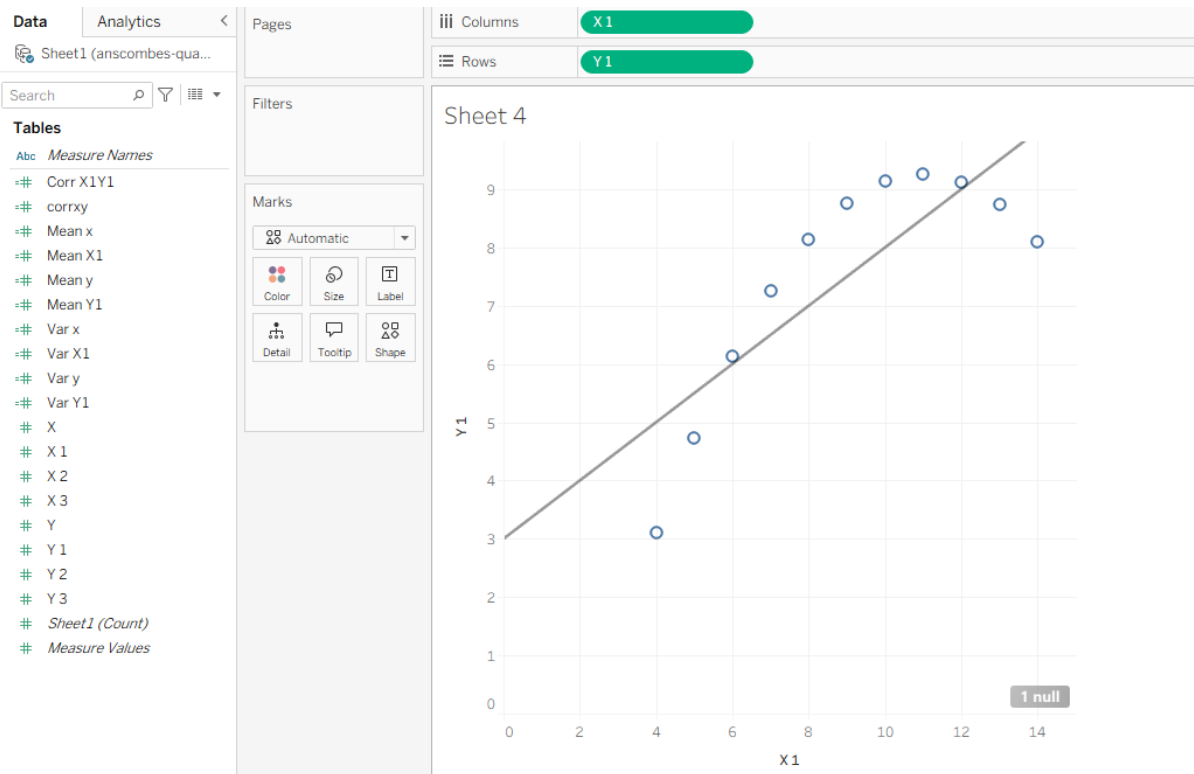
Columns

Rows

Measure Names

Sheet 3

Corr X1Y1	0.82
Mean X1	9.00
Mean Y1	7.50
Var X1	11.00
Var Y1	4.13



← → ↺ ↻
📄 📊
📈 📉
🔍 ⏏️ ⏴ ⏵
🖋️ 📎 📄 ⭐
Standard

Data

Analytics

Sheet1 (anscombes-qua...)

🔍
🔍
🔍

Tables

Abc
Measure Names

Corr X1Y1
Corr X2Y2
corrxy
Mean x
Mean X1
Mean X2
Mean y
Mean Y1
Mean Y2
Var x
Var X1
Var X2
Var y
Var Y1
Var Y2
X
X 1
X 2
X 3

Pages

Filters

Measure Names

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

AGG(Corr X2Y2)

AGG(Mean Y2)

AGG(Var X2)

AGG(Var Y2)

SUM(X 2)

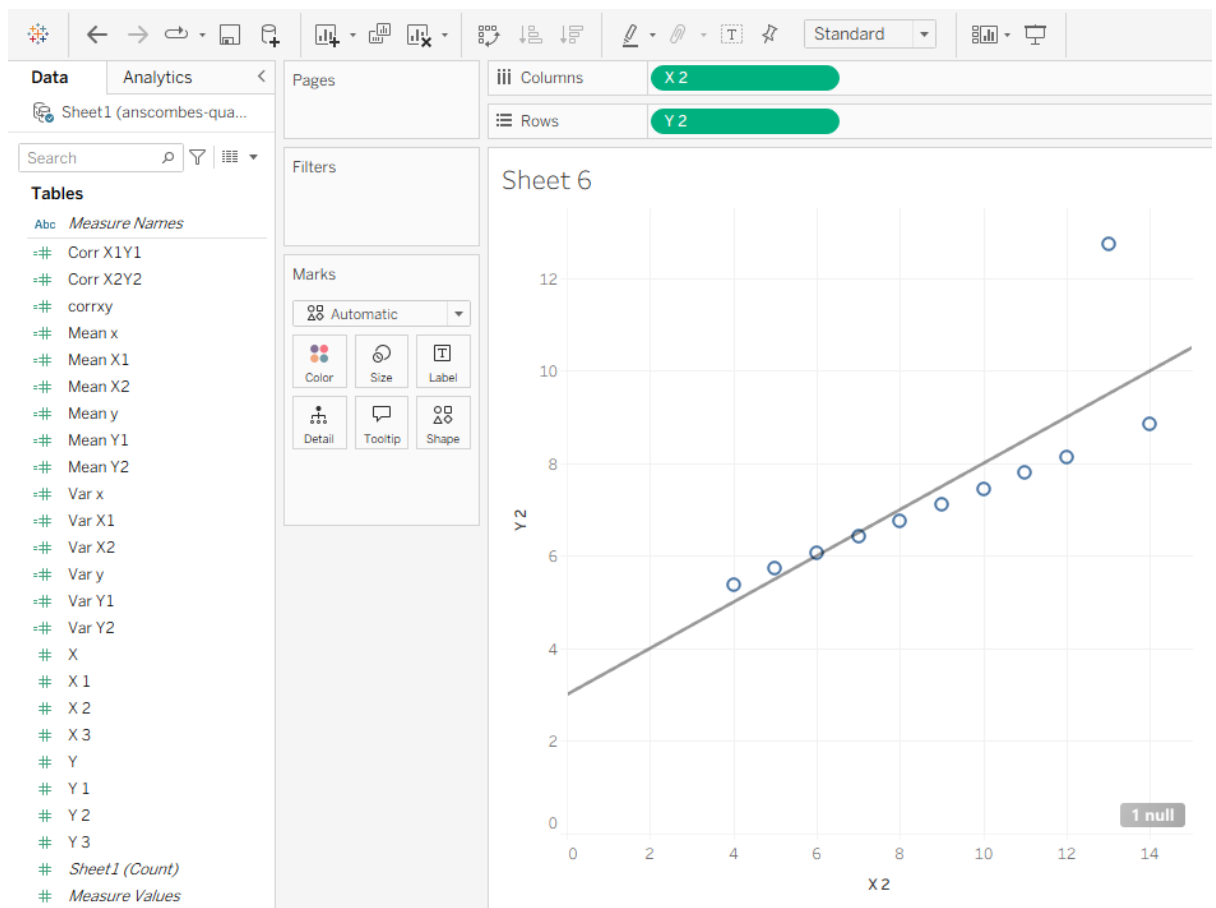
Columns

Rows

Measure Names

Sheet 5

Corr X2Y2	0.82
Mean Y2	7.50
Var X2	11.00
Var Y2	4.12
X 2	99.00



Data

Analytics

Sheet1 (anscombes-qua...

Search

Tables

-# Mean x

-# Mean X1

-# Mean X2

-# Mean X3

-# Mean y

-# Mean Y1

-# Mean Y2

-# Mean Y3

-# Var x

-# Var X1

-# Var X2

-# Var X3

-# Var y

-# Var Y1

-# Var Y2

-# Var Y3

X

X 1

X 2

X 3

Y

Y 1

Pages

Filters

Measure Names

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

AGG(Corr X3Y3)

AGG(Mean X3)

AGG(Mean Y3)

AGG(Var X3)

AGG(Var Y3)

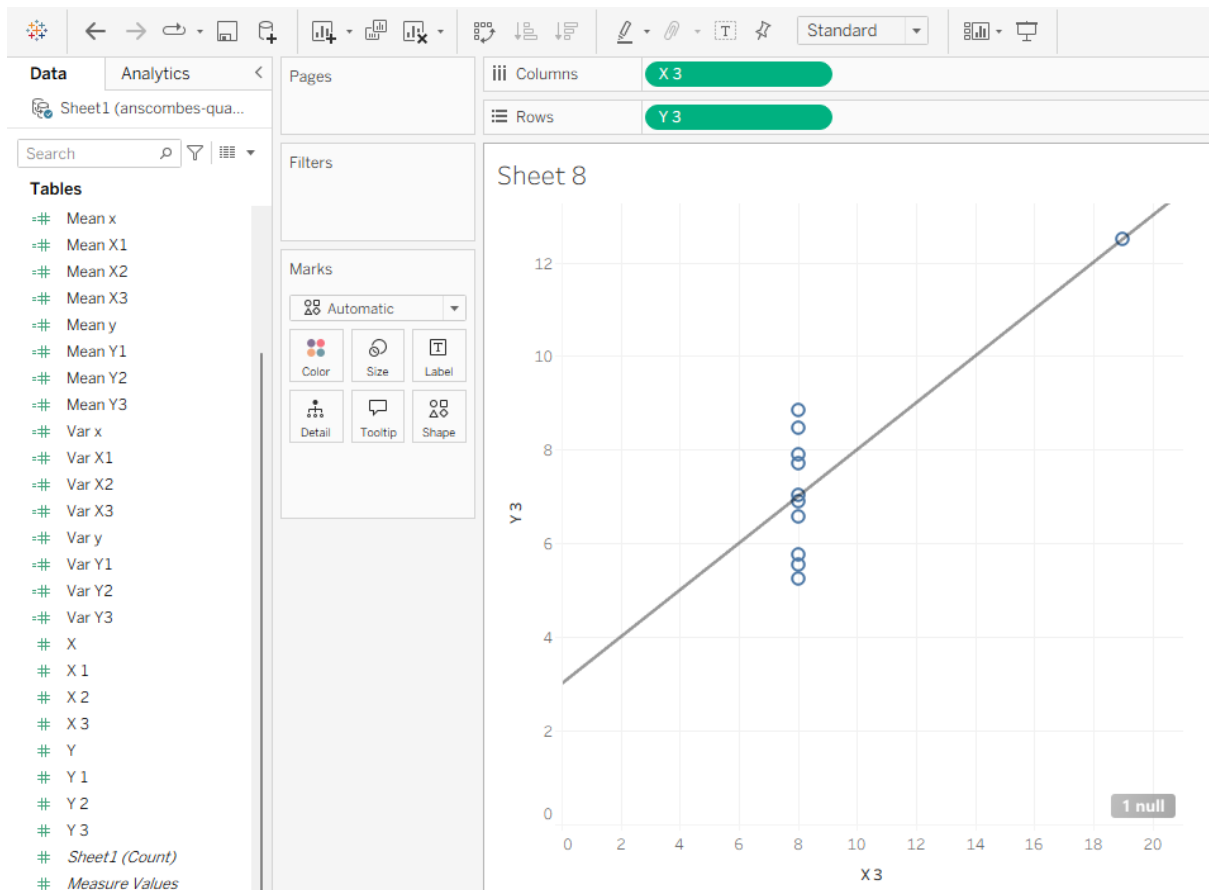
Columns

Rows

Measure Names

Sheet 7

Corr X3Y3	0.82
Mean X3	9.00
Mean Y3	7.50
Var X3	11.00
Var Y3	4.12



Pb3).

Data | Analytics | Pages | Columns: Measure Names | Rows: Category, Sub-Category

Search: Orders (Global.Supersto...)

Tables

- Category
 - Category
 - Sub-Category
- City
- Country
- Customer ID
- Customer Name
- Market
- Order Date
- Order ID
- Order Priority
- Postal Code
- Product ID
- Product Name
- Region
- Row ID
- Segment
- Ship Date
- Ship Mode
- State
- Measure Names

Filters

Measure Names: Sal..

Marks

Automatic

Color, Size, Text

Detail, Tooltip

Measure Values

SUM(Sales)

1.Category Sub Category Hierarchy

Category	Sub-Catego..	Sales
Furniture	Bookcases	1,466,572
	Chairs	1,501,682
	Furnishings	385,578
	Tables	757,042
Office Supplies	Appliances	1,011,064
	Art	372,092
	Binders	461,912
	Envelopes	170,904
	Fasteners	83,242
	Labels	73,404
	Paper	244,292
	Storage	1,127,086
	Supplies	243,074
Technology	Accessories	749,237
	Copiers	1,509,436
	Machines	779,060
	Phones	1,706,824

b)

Data | Analytics | Pages | Columns: Measure Names | Rows: Country, State, City

Search: Orders (Global.Supersto...)

Tables

- Category
 - Category
 - Sub-Category
- Country
 - Country
 - State
 - City
- Customer ID
- Customer Name
- Market
- Order Date
- Order ID
- Order Priority
- Postal Code
- Product ID
- Product Name
- Region
- Row ID
- Segment
- Ship Date
- Ship Mode
- Measure Names
- Discount
- Profit
- Quantity
- Sales

Filters

Measure Names

Marks

Automatic

Color, Size, Text

Detail, Tooltip

Measure Values

SUM(Sales)

SUM(Shipping Cost)

2.Country state city hierarchy

Country	State	City	Sales	Shippin..
Afghanistan	Hirat	Herat	1,695	89
	Kabul	Kabul	16,951	2,034
	Kandahar	Kandahar	1,867	90
	Nangarhar	Jalalabad	1,160	87
Albania	Durrës	Durres	612	212
	Elbasan	Elbasan	1,619	302
	Korçë	Korce	77	6
	Shkodër	Shkoder	593	107
Algeria	Vlorë	Vlore	986	72
	Alger	Algiers	9,270	1,198
	Annaba	Annaba	2,921	362
	Batna	Barika	239	57
		Batna	1,733	121
	Bechar	Bechar	150	14
	Bejaia	Bejaia	632	42
	Constantine	Constantine	1,574	159
	Djelfa	Ain Oussera	238	25
		Dar Chioukh	26	1
		Messaad	1,254	57
	Guelma	Guelma	3,665	372
	Laghouat	Laghouat	799	88
	Medea	Medea	123	20
	Mostaganem	Mostaganem	440	43
	Oran	Oran	7,337	671
	Saida	Saida	3,524	339
	Sidi Bel Abbes	Sidi Bel Abbes	569	32

c)

Data

Analytics

Orders (Global.Supersto...

Search

Tables

Category

Category

Sub-Category

Country

Country

State

City

Customer ID

Customer Name

Market

Order Date

Order ID

Order Priority

Postal Code

Product ID

Product Name

Region

Row ID

Segment

Ship Date

Ship Mode

Sub-Category Set

Measure Names

Pages

Filters

Measure Names: Sal..

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Measure Values

Measure Values

SUM(Sales)

Columns

Measure Names

Rows

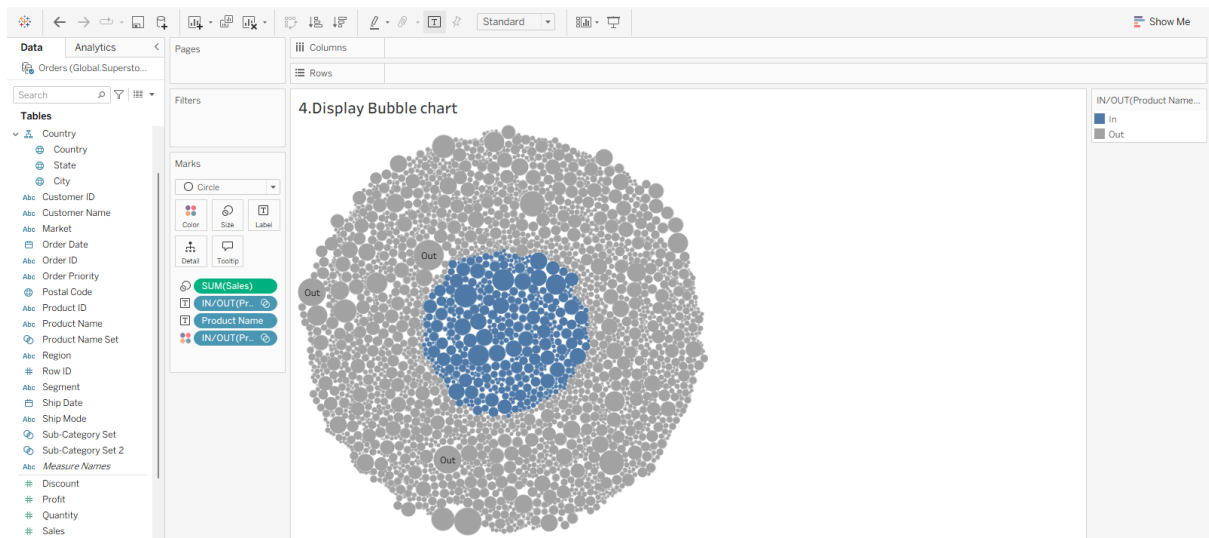
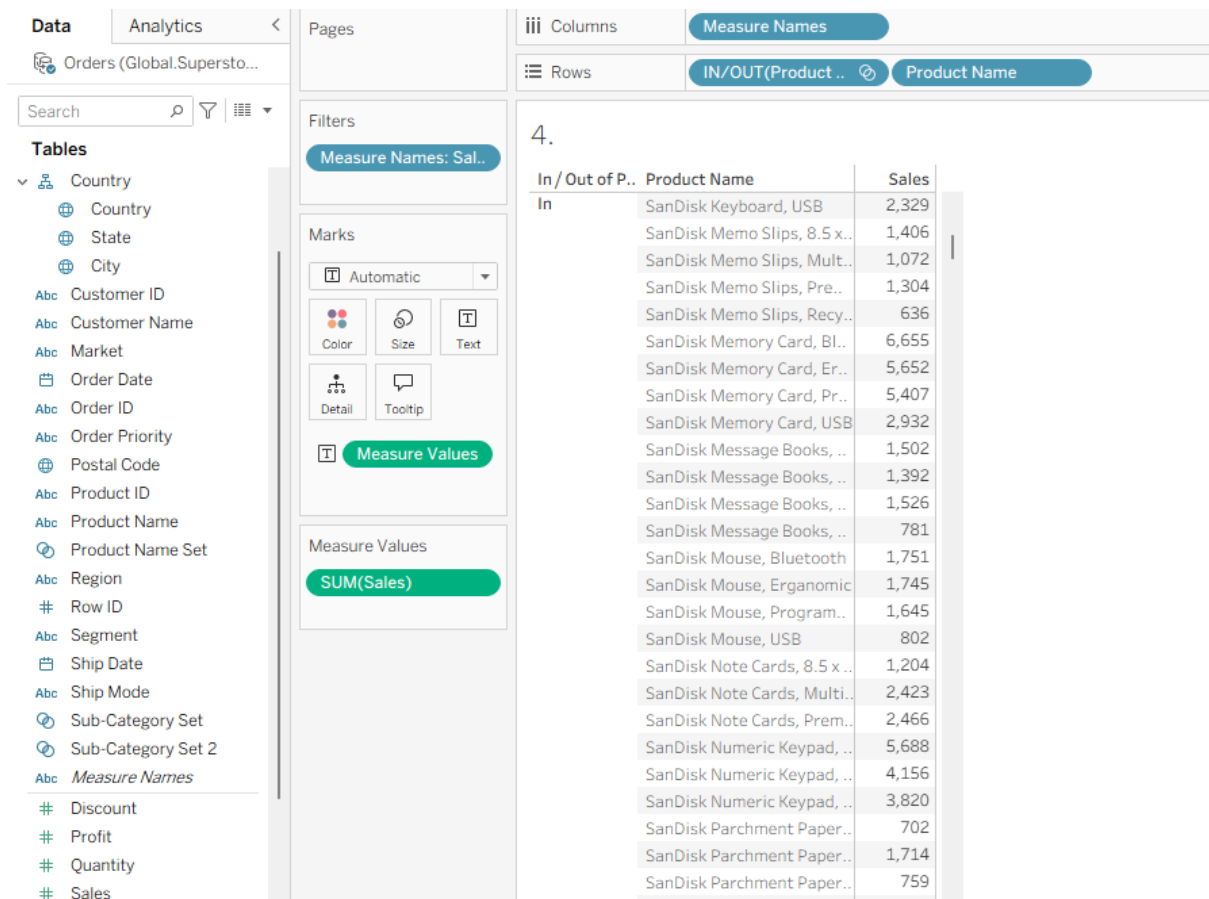
Sub-Category

IN/OUT(Sub-Cate..

Sheet 3

Sub-Catego..	In / Out of S..	Sales
Accessories	In	749,237
Appliances	In	1,011,064
Art	In	372,092
Binders	In	461,912
Bookcases	In	1,466,572
Chairs	In	1,501,682
Copiers	In	1,509,436
Envelopes	In	170,904
Fasteners	In	83,242
Furnishings	In	385,578
Labels	In	73,404
Machines	In	779,060
Paper	In	244,292
Phones	In	1,706,824
Storage	In	1,127,086
Supplies	In	243,074
Tables	In	757,042

d)



e)

P4.

A)

×

```
IF [Basic Salary]<20000 THEN
[Basic Salary]*0.10
ELSE
[Basic Salary]*0.12
END
```

The calculation is valid.

2 Dependencies ▼

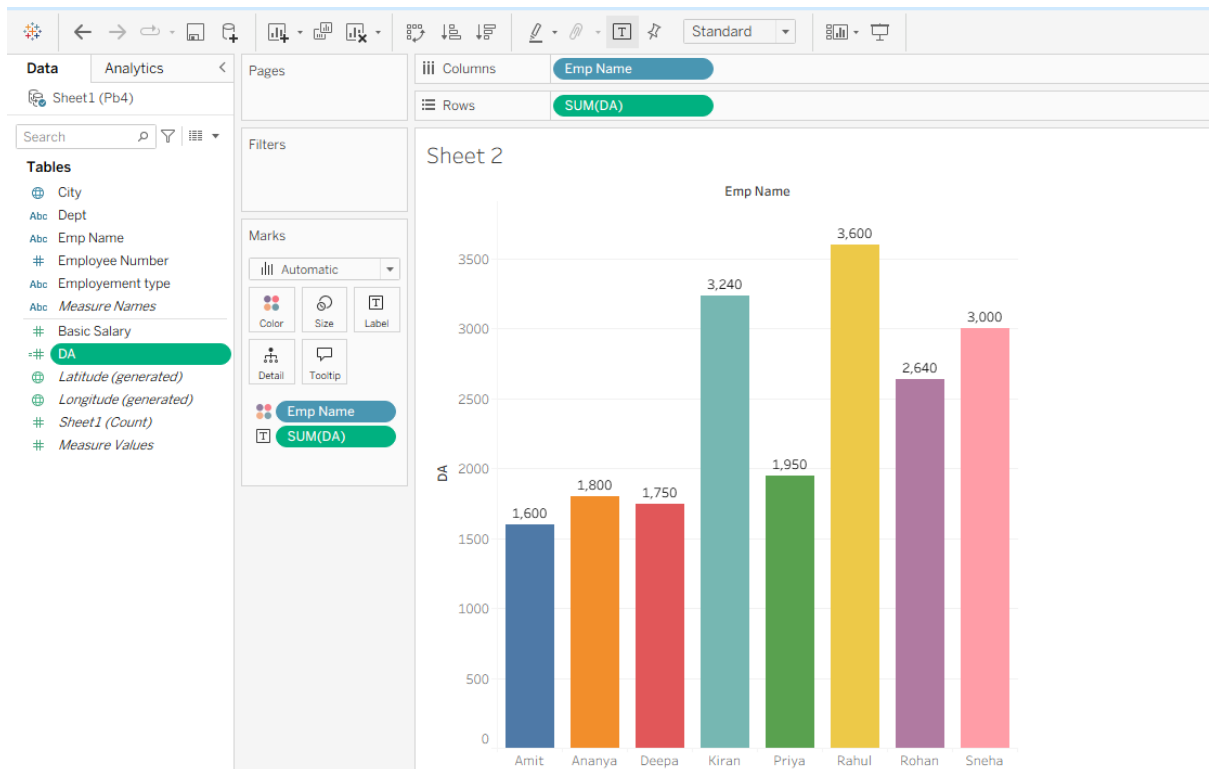
Apply

OK

Columns	
Rows	Emp Name

Sheet 1

Emp Name	
Amit	1,600
Ananya	1,800
Deepa	1,750
Kiran	3,240
Priya	1,950
Rahul	3,600
Rohan	2,640
Sneha	3,000



B)

HR



```
IF LOWER([City])="metro" THEN
[Basic Salary]*0.20
ELSE
[Basic Salary]*0.15
END
```

The calculation is valid.

Apply

OK

Data Analytics <

Sheet1 (Pb4)

Search 🔍

Tables

- City
- Dept
- Emp Name
- Employee Number
- Employement type
- Measure Names
- Basic Salary**
- DA
- HR
- Latitude (generated)
- Longitude (generated)
- Sheet1 (Count)
- Measure Values

Pages

Columns

Rows Emp Name

HRA

Emp Name	
Amit	2,400
Ananya	3,600
Deepa	2,625
Kiran	5,400
Priya	3,900
Rahul	6,000
Rohan	3,300
Sneha	5,000

Filters

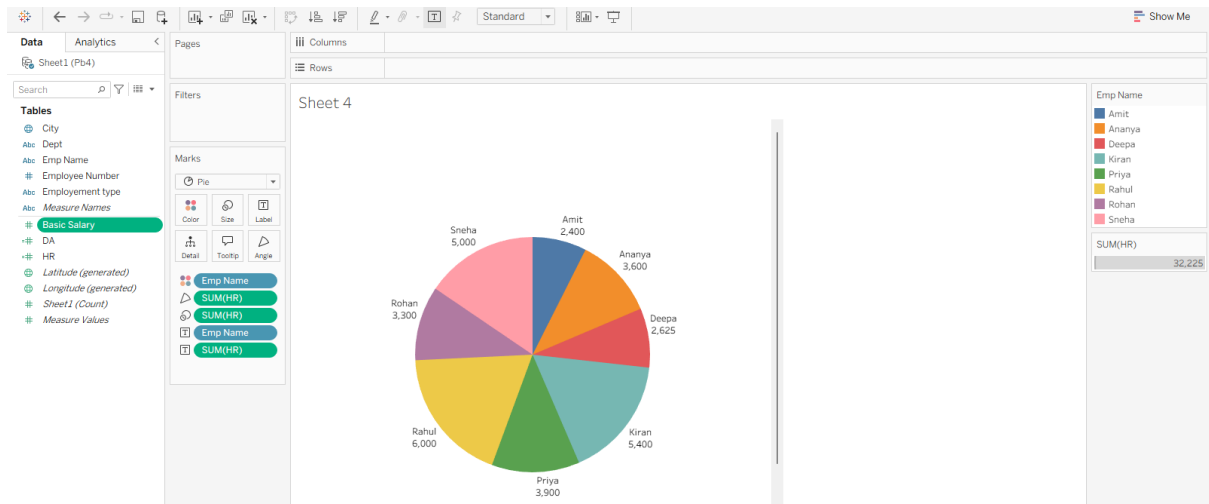
Marks

Automatic

Color Size Text

Detail Tooltip

SUM(HR)



C)

Sheet1 (Pb4)

Measure Names

Emp Name

Employement type

PF

Emp Name	Employeme..	PF
Amit	Permanenet	1,920
Ananya	Permanenet	2,160
Deepa	Permanenet	2,100
Kiran	Permanenet	3,240
Priya	Contract	0
Rahul	Contract	0
Rohan	Permanenet	2,640
Sneha	Contract	0

Measure Values

SUM(PF)

D)

Data | Analytics | Pages | Columns: Measure Names | Rows: Emp Name

Sheet1 (Pb4)

Search

Tables

- City
- Dept
- Emp Name
- Employee Number
- Employment type
- Measure Names
- Basic Salary
- DA
- Gross Salary
- HR
- Net salary
- PF
- Latitude (generated)
- Longitude (generated)
- Sheet1 (Count)
- Measure Values

Filters

Measure Names

Marks

Automatic

Color Size Text

Detail Tooltip

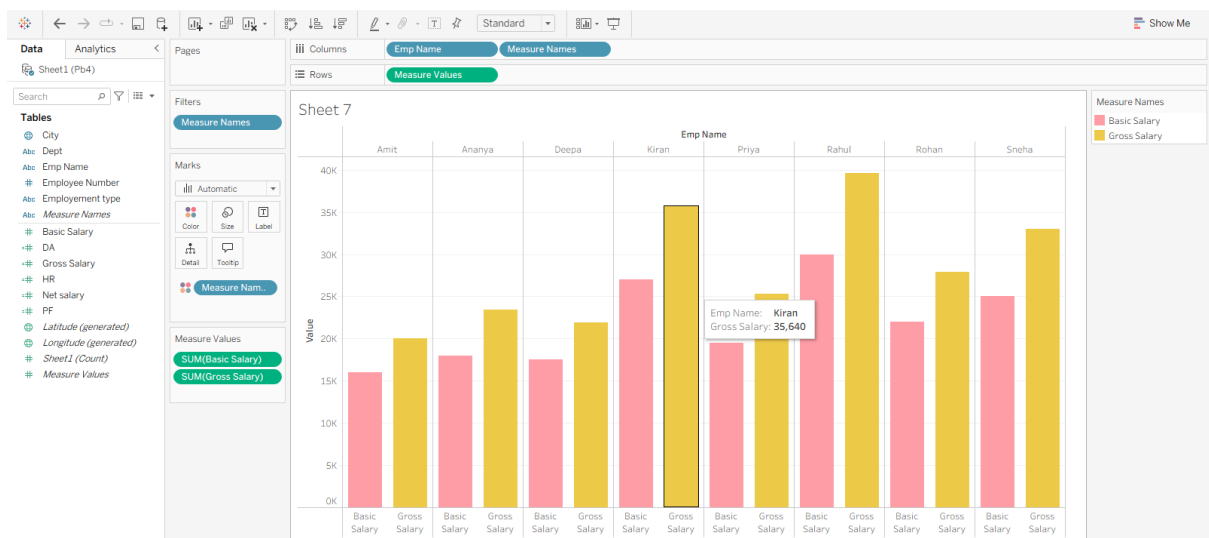
Measure Values

SUM(Basic Salary)

SUM(Gross Salary)

Gross Salary

Emp Name	Basic Salary	Gross Salary
Amit	16,000	20,000
Ananya	18,000	23,400
Deepa	17,500	21,875
Kiran	27,000	35,640
Priya	19,500	25,350
Rahul	30,000	39,600
Rohan	22,000	27,940
Sneha	25,000	33,000



E)

E)

Data | Analytics | Pages | Columns: Measure Names | Rows: Emp Name

Sheet1 (Pb4)

Search

Tables

- City
- Dept
- Emp Name
- Employee Number
- Employment type
- Measure Names
- Basic Salary
- DA
- Gross Salary
- HR
- Net salary
- PF
- Latitude (generated)
- Longitude (generated)
- Sheet1 (Count)
- Measure Values

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

SUM(Net salar..

Sheet 8

Emp Name	No Meas ure..
Amit	18,080
Ananya	21,240
Deepa	19,775
Kiran	32,400
Priya	25,350
Rahul	39,600
Rohan	25,300
Sneha	33,000

Data | Analytics | Pages | Columns: | Rows: |

Sheet1 (Pb4)

Search

Tables

- City
- Dept
- Emp Name
- Employee Number
- Employment type
- Measure Names
- Basic Salary
- DA
- Gross Salary
- HR
- Net salary
- PF
- Latitude (generated)
- Longitude (generated)
- Sheet1 (Count)
- Measure Values

Filters

Marks

Circle

Color Size Label

Detail Tooltip

SUM(Net salar.. Emp Name SUM(Net salar.. Emp Name

Sheet 9

Emp Name

- Amit
- Ananya
- Deepa
- Kiran
- Priya
- Rahul
- Rohan
- Sneha

Pb5)

A.

Pages	Columns	YEAR(Order Date)
Filters	Rows	Sub-Category
Marks	A.Display total profit trends over the year for each sun-category	
Automatic	Order Date	
Color	Sub-Catego..	2013 2014 2015 2016
Size	Accessories	15,720 33,507 38,805 41,594
Text	Appliances	22,838 24,316 41,486 53,041
Detail	Art	10,399 11,841 15,630 20,084
Tooltip	Binders	11,447 17,123 21,524 22,356
SUM(Profit)	Bookcases	27,519 28,137 43,049 63,219
	Chairs	29,943 28,755 40,449 41,248
	Copiers	30,375 51,843 72,301 104,049
	Envelopes	5,066 6,603 9,572 8,360
	Fasteners	1,937 3,004 2,912 3,672
	Furnishings	7,310 9,662 14,605 15,391
	Labels	3,119 3,230 3,607 5,055
	Machines	9,225 15,404 15,612 18,628
	Paper	10,665 10,687 16,880 20,976
	Phones	53,927 45,223 46,909 70,658
	Storage	15,944 21,489 32,011 39,017
	Supplies	4,580 5,012 5,625 7,365
	Tables	-11,075 -8,422 -14,040 -30,546

b)

Pages	Columns	Measure Names
	Rows	Country State

Filters

Measure Names: Sal..

Marks

Automatic

Color Size Text

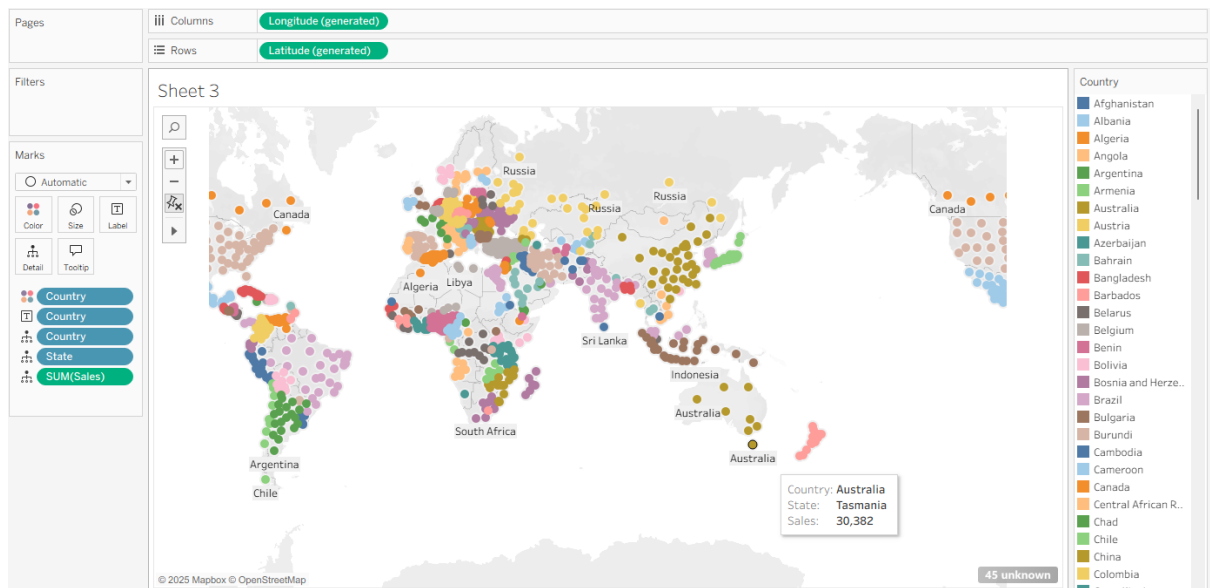
Detail Tooltip

Measure Values

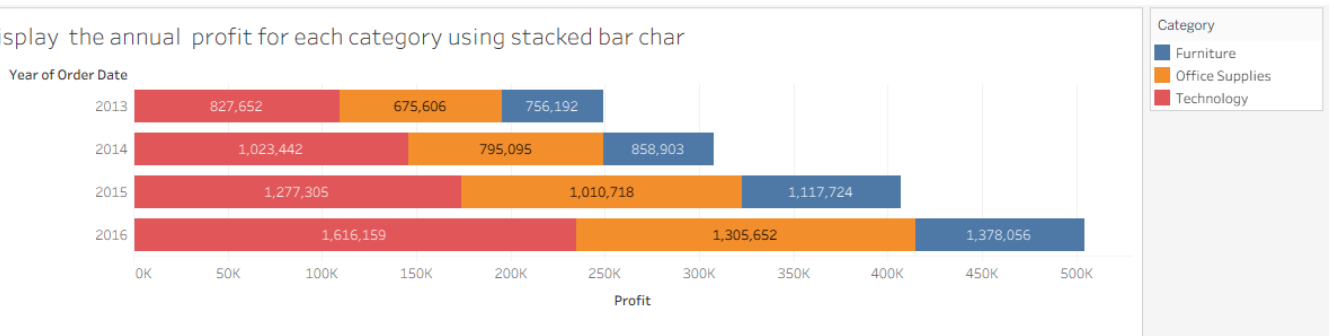
SUM(Sales)

Sheet 2

Country	State	Sales
Afghanistan	Hirat	1,695
	Kabul	16,951
	Kandahar	1,867
	Nangarhar	1,160
Albania	Durrës	612
	Elbasan	1,619
	Korçë	77
	Shkodër	593
Algeria	Vlorë	986
	Alger	9,270
	Annaba	2,921
	Batna	1,973
	Bechar	150
	Bejaia	632
	Constantine	1,574
	Djelfa	1,519
	Guelma	3,665
	Laghouat	799
	Medea	123
	Mostaganem	440
	Oran	7,337
	Saida	3,524
	Sidi Bel Abbes	569
	Skikda	635
	Tipaza	678
	Tizi Ouzou	198



c. Display the annual profit for each category using stacked bar chart



c)

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

SUM(Profit)

Columns

YEAR(Order Date)

Rows

Category

Sheet 4

	Order Date			
Category	2013	2014	2015	2016
Furniture	53,697	58,133	84,063	89,312
Office Supplies	85,997	103,306	149,246	179,926
Technology	109,247	145,977	173,627	234,928

D)

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Tableau Desktop Public Edition

Orders (Global.Supersto...)

Search

Tables

- Category
- City
- Country
- Customer ID
- Customer Name
- Market
- Order Date
- Order ID
- Order Priority
- Postal Code
- Product ID
- Product Name
- Region
- Row ID
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Measure Names
- Discount
- Profit
- Quantity
- Sales
- Shipping Cost
- Latitude (generated)
- Longitude (generated)

Columns: YEAR(Or)

Rows:

Filters:

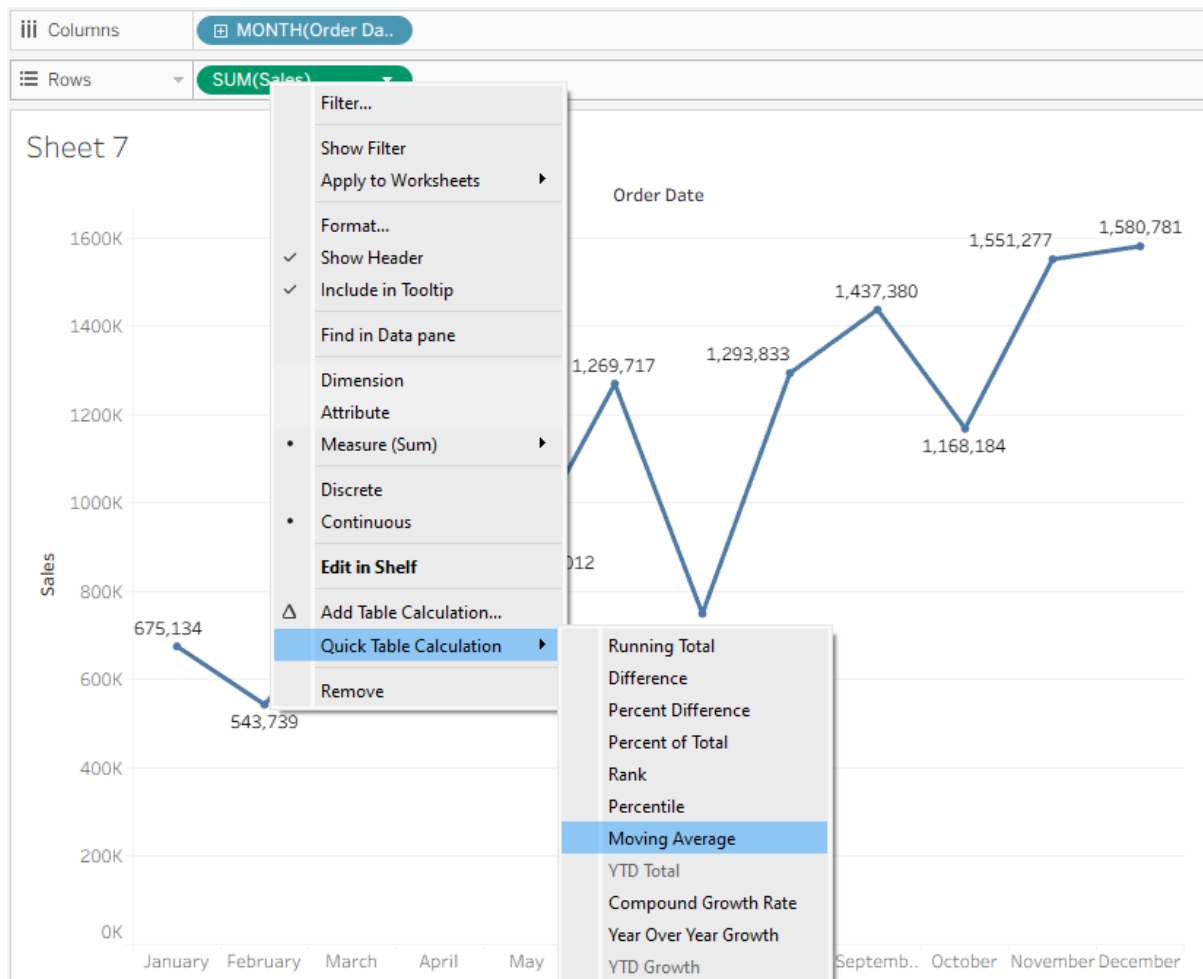
Marks: Automatic

Color Size Text Detail Tooltip

Sheet 7

	Order Date
2013	2014 2015
Abc	Abc Abc

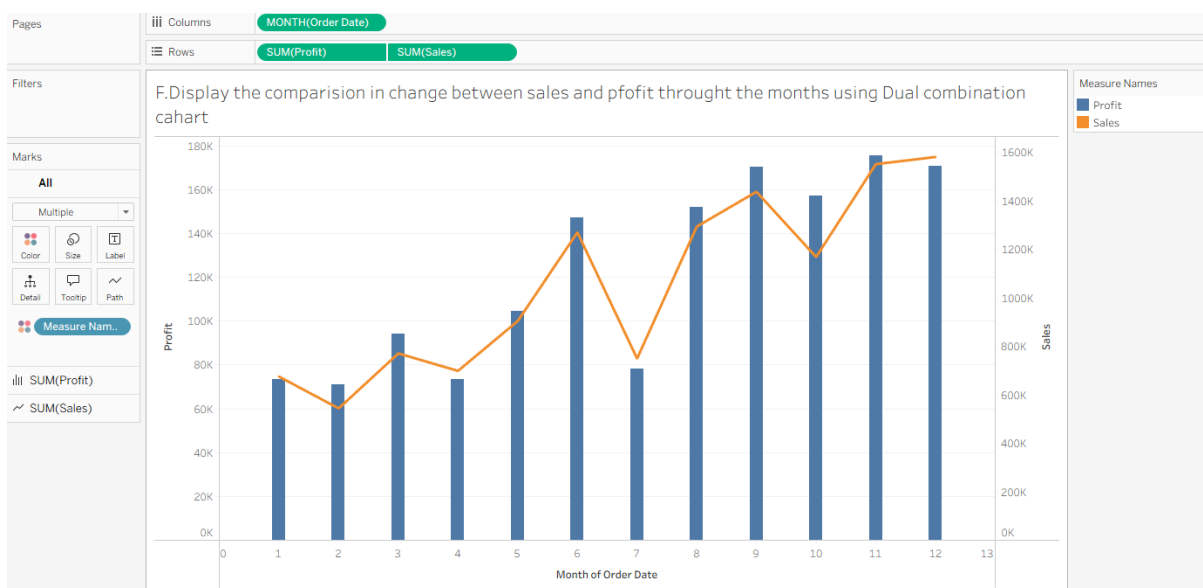
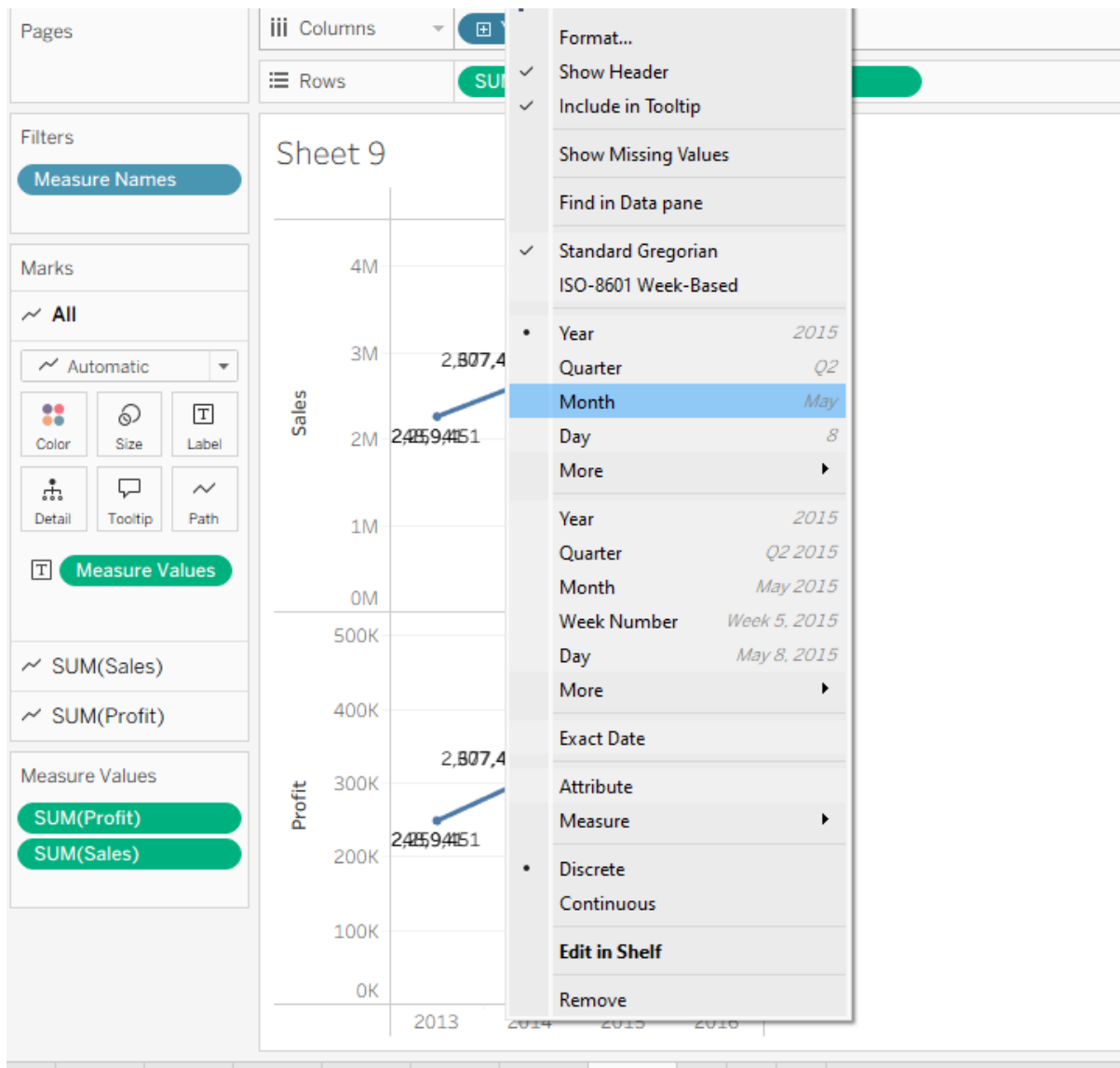
Show Filter
 Show Highlighter
 Apply to Worksheets
 Sort...
 Format...
 Show Header
 Include in Tooltip
 Show Missing Values
 Find in Data pane
 Standard Gregorian
 ISO-8601 Week-Based
 Year 2015
 Quarter Q2
Month May
 Day 8
 More
 Year 2015
 Quarter Q2 2015
 Month May 2015
 Week Number Week 5, 2015
 Day May 8, 2015
 More
 Exact Date
 Attribute
 Measure
 Discrete
 Continuous
 Edit in Shelf
 Remove

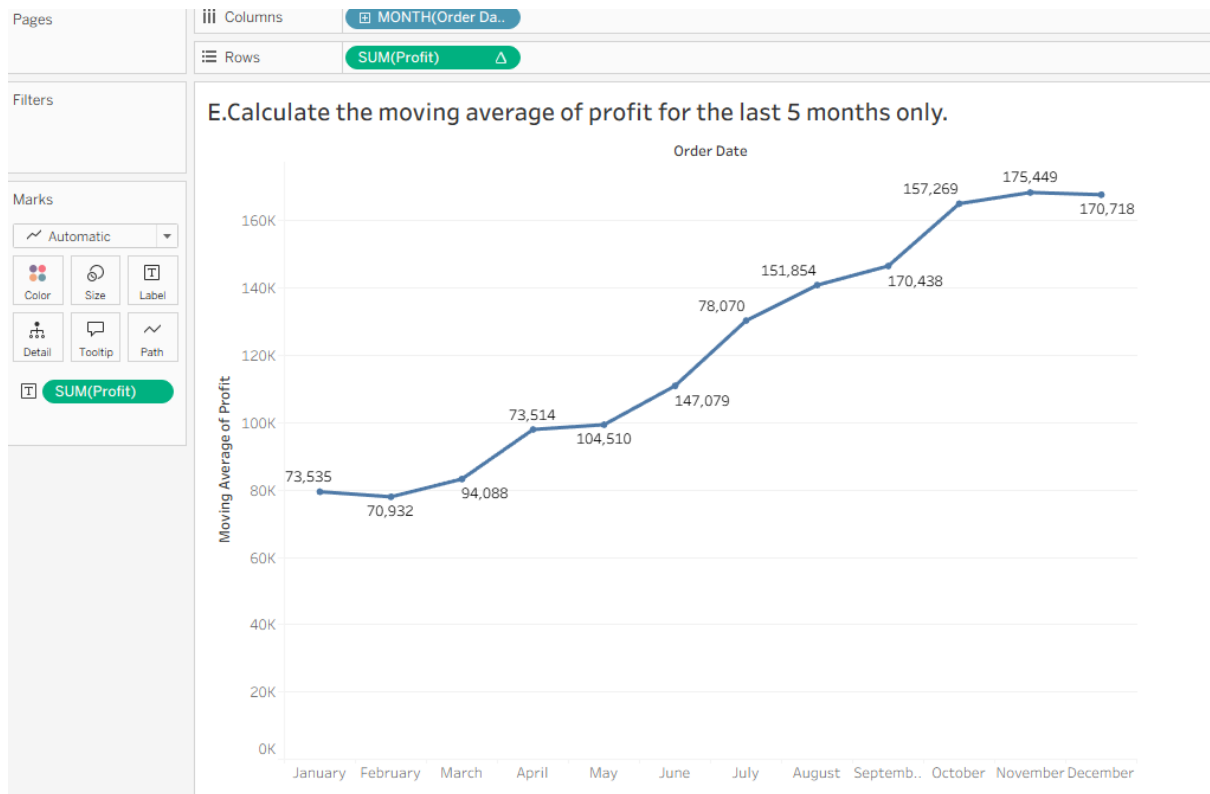


D. Calculate the moving average of sales for the last 3 months only



E.





F.

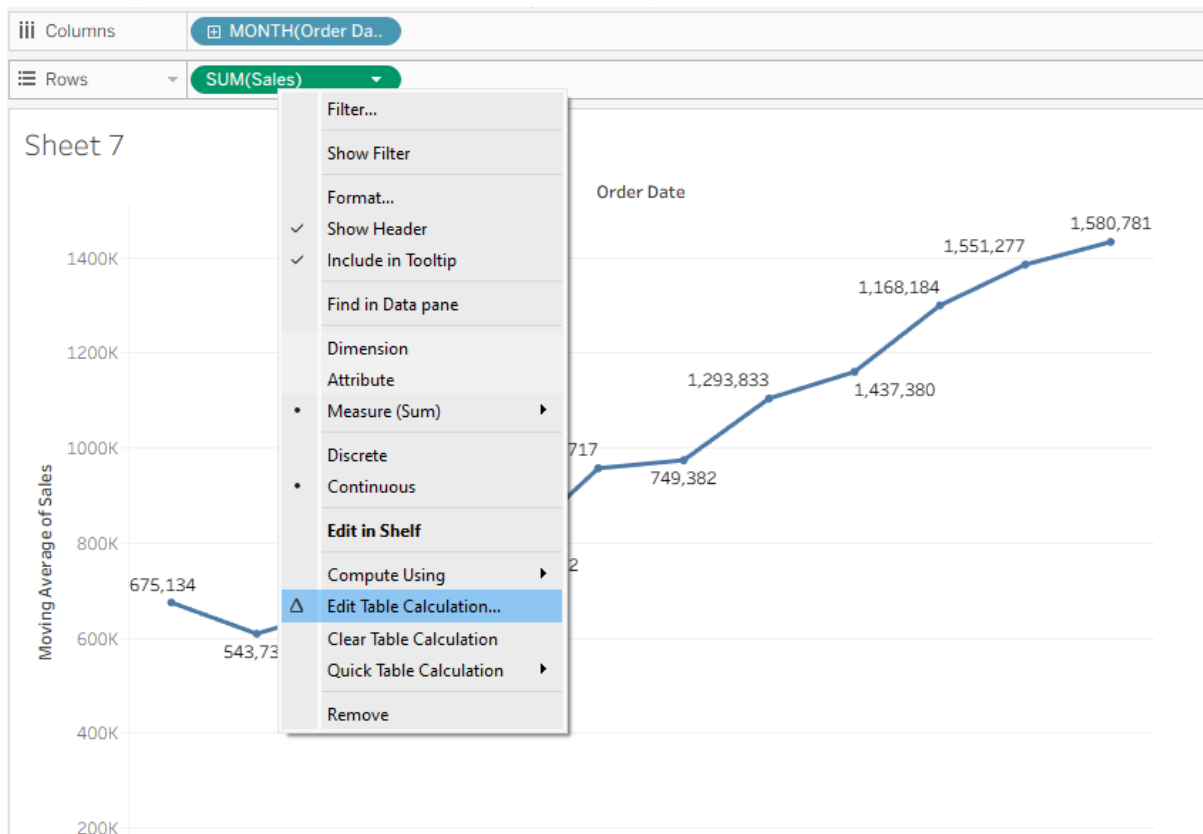


Table Calculation

Moving Average of Sales

Calculation Type

Moving Calculation

Average, prev 1, next 1

Compute Using

Table (across)

Cell

Specific Dimensions

☒ Month of Order Date

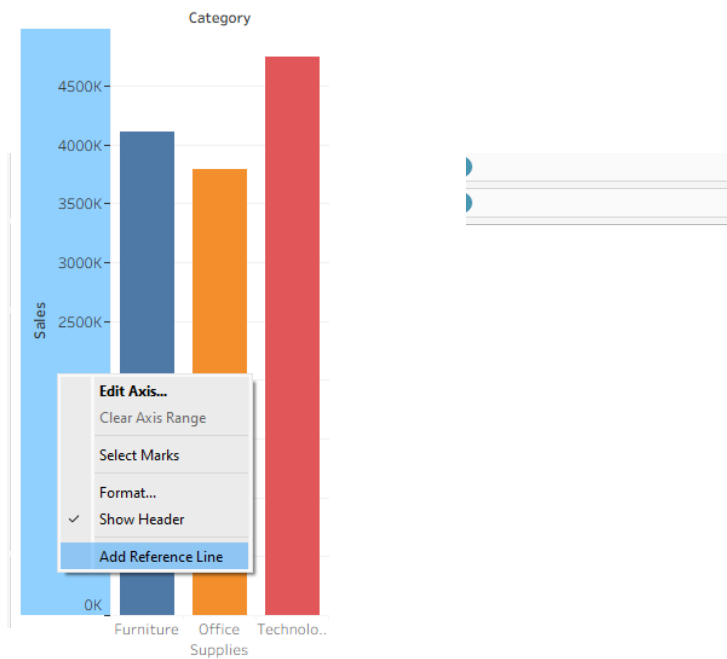
☐ Add secondary calculation

☒ Show calculation assistance

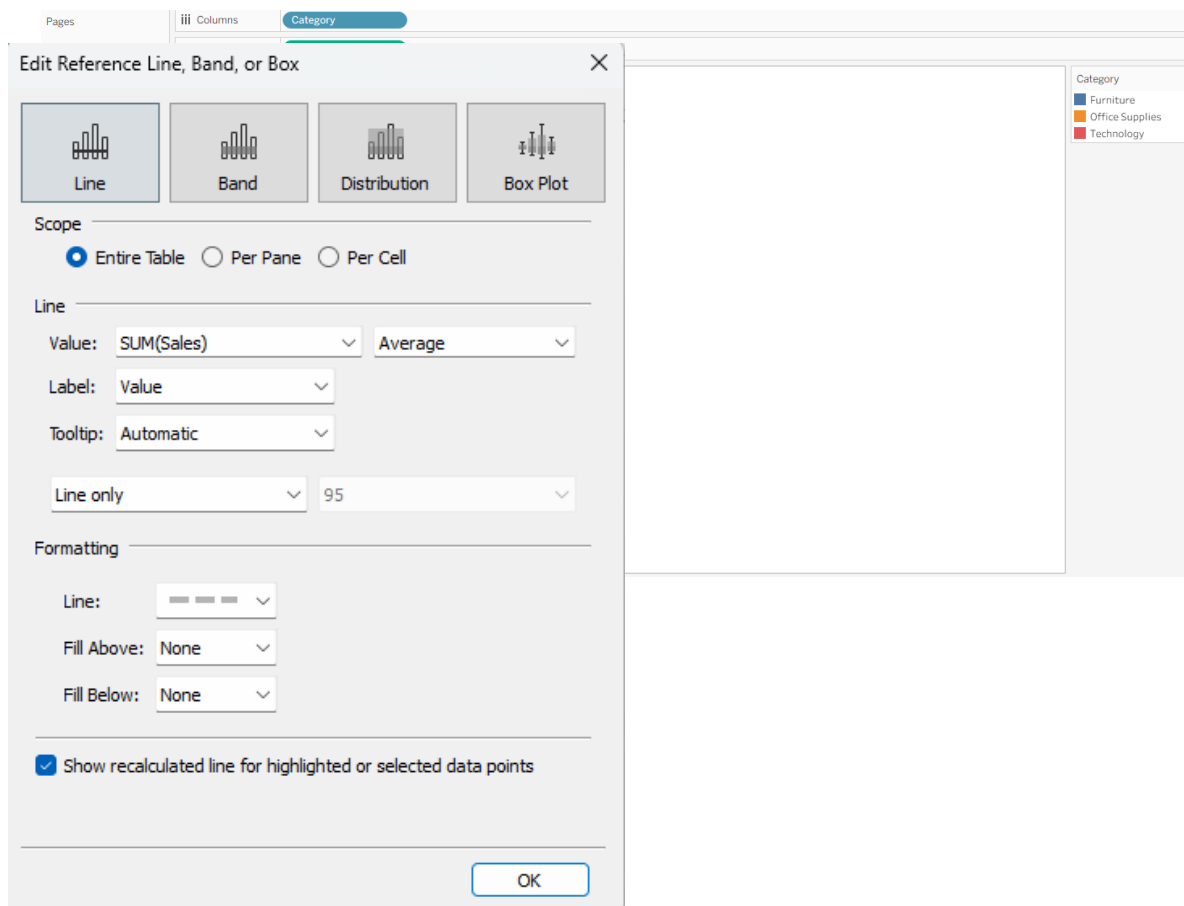


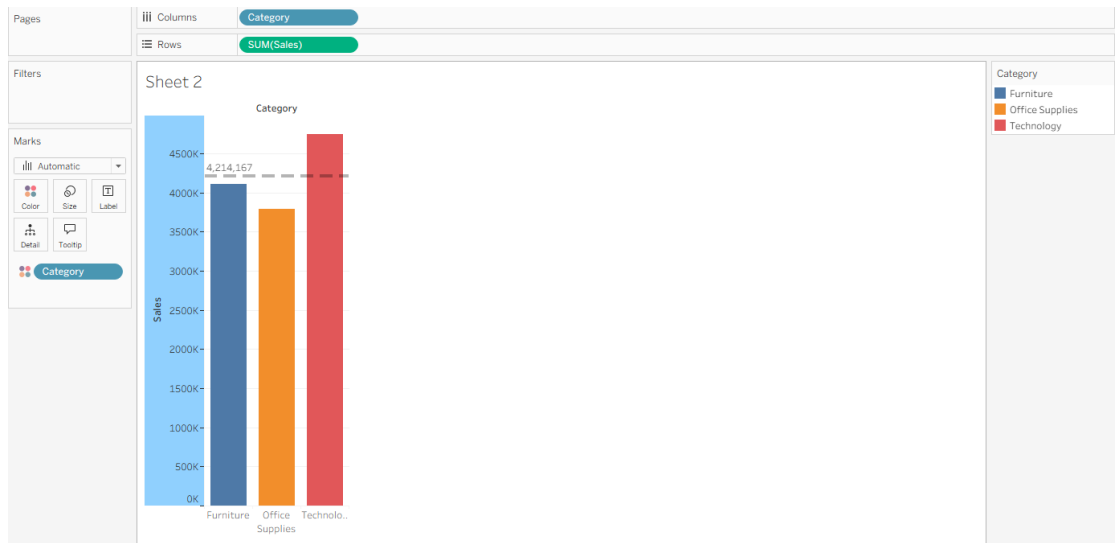
6. Consider global superstore data and perform reference line Operations.

Sheet 2



a) Display reference line per table for average sales under category





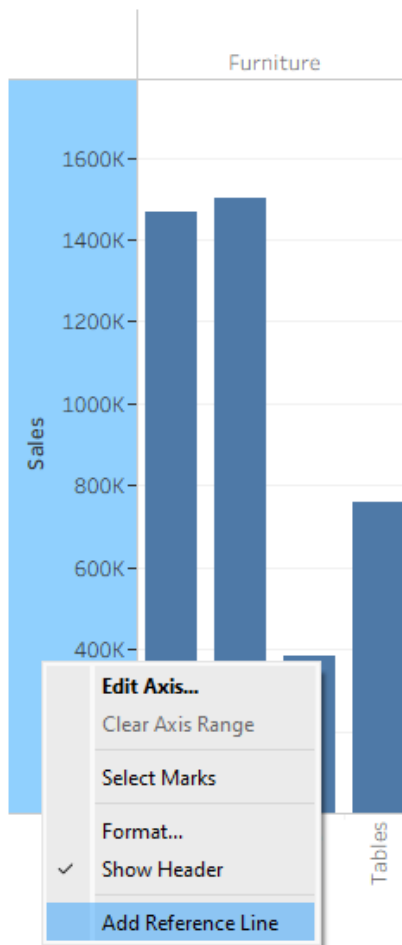
b) Display Reference line per table for average Sales under category and subcategory

Columns	Measure Names	
Rows	Category	Sub-Category

Sheet 3

Category	Sub-Catego..	No Measure Value
Furniture	Bookcases	1,466,572
	Chairs	1,501,682
	Furnishings	385,578
	Tables	757,042
Office Supplies	Appliances	1,011,064
	Art	372,092
	Binders	461,912
	Envelopes	170,904
	Fasteners	83,242
	Labels	73,404
	Paper	244,292
	Storage	1,127,086
	Supplies	243,074
Technology	Accessories	749,237
	Copiers	1,509,436
	Machines	779,060
	Phones	1,706,824

Sheet 3



Pages

Filters

Marks

Automatic

Color
Size
Text

Detail
Tooltip

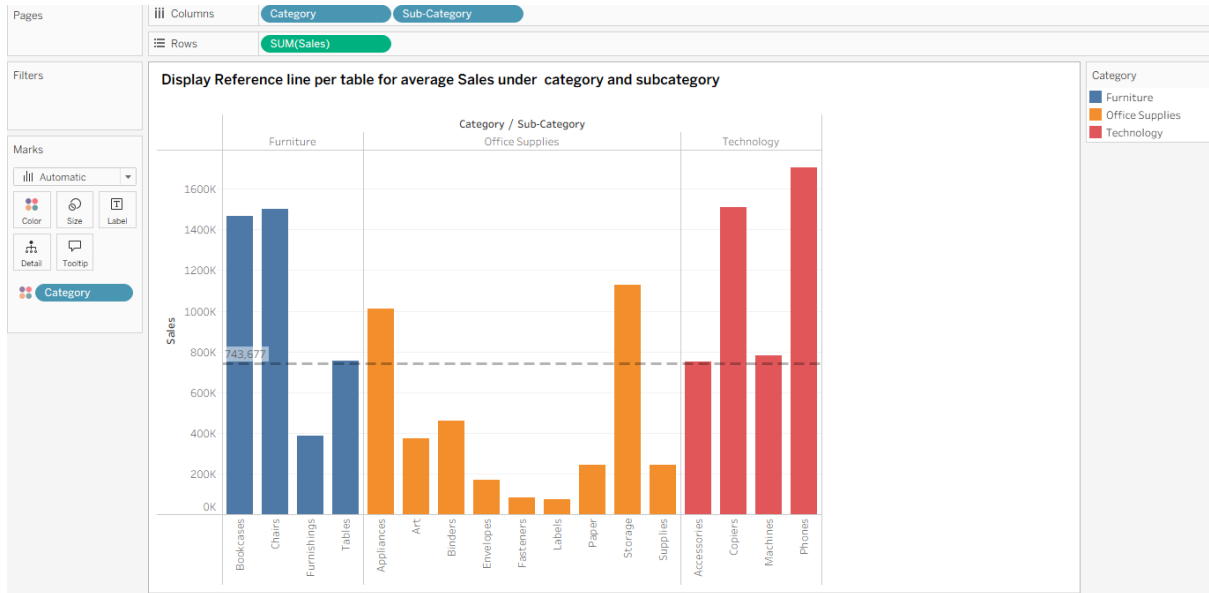
SUM(Sales)

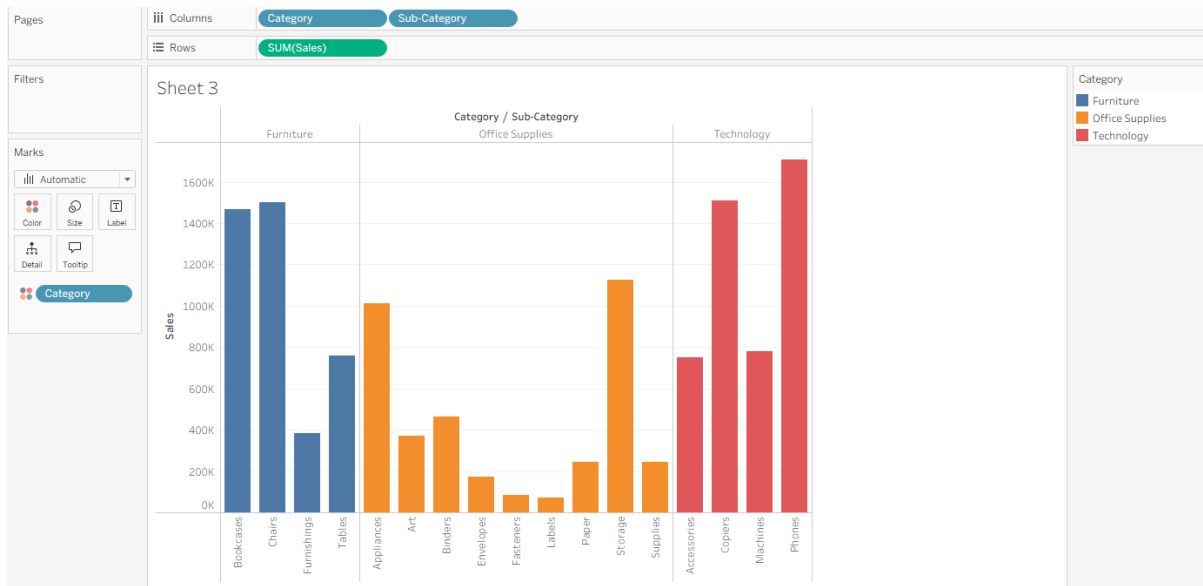
Columns
Measure Names

Rows
Category
Sub-Category

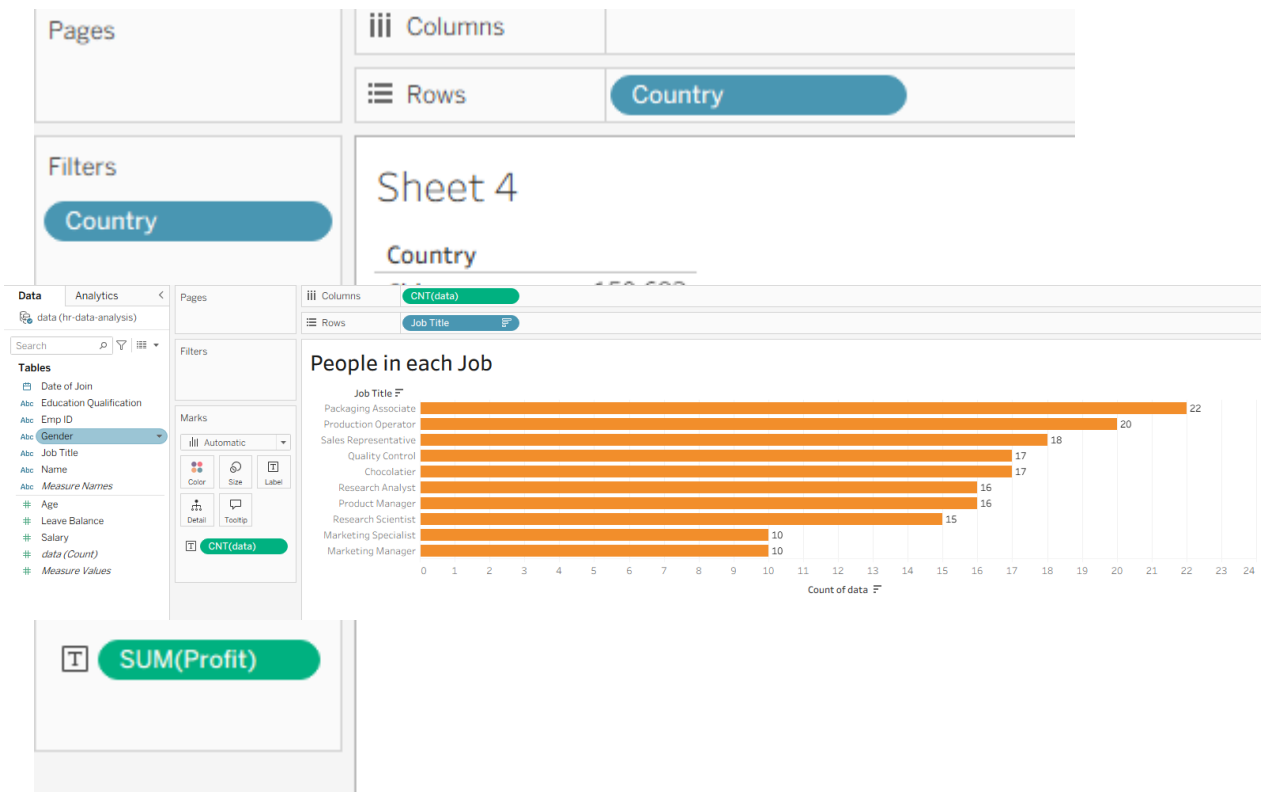
Sheet 2

Category	Sub-Catego..	No Measure Value
Furniture	Bookcases	1,466,572
	Chairs	1,501,682
	Furnishings	385,578
	Tables	757,042
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Technology	Accessories	749,237
	Copiers	1,509,436
	Machines	779,060
	Phones	1,706,824





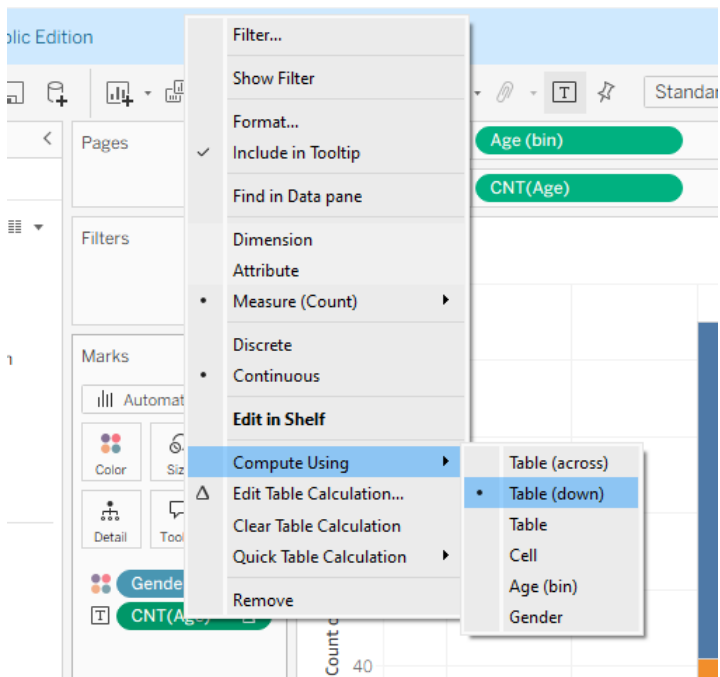
d)



Part C

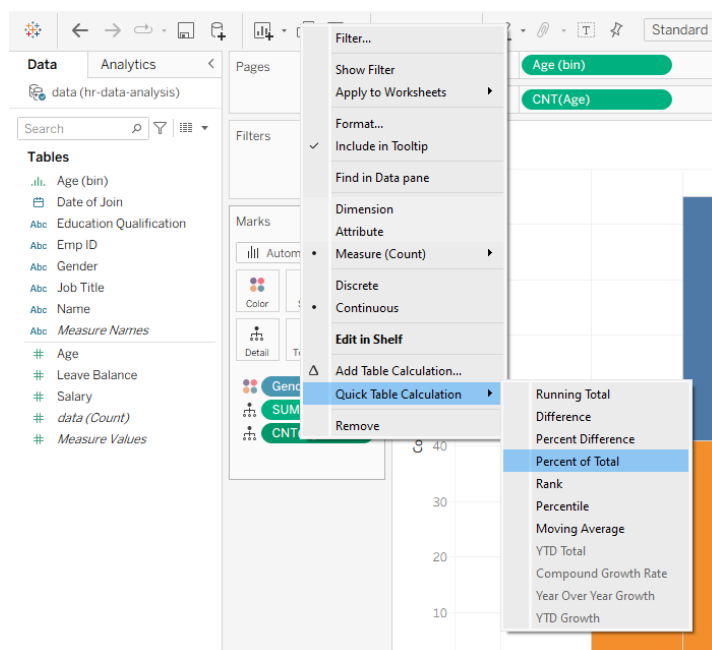
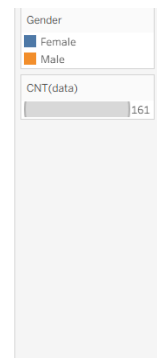
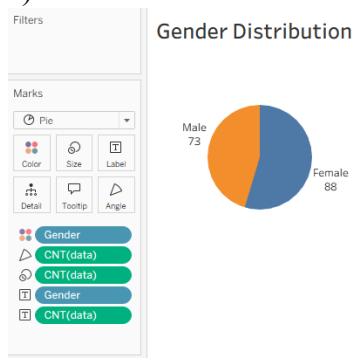
14. Create given HR dashboard using HR data.

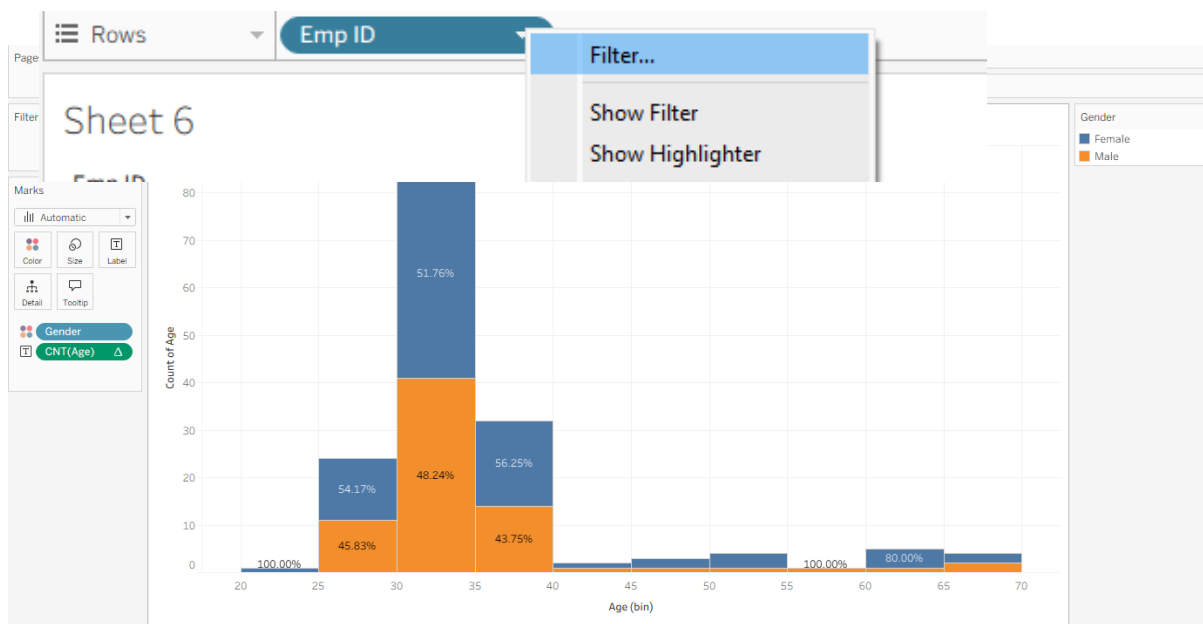
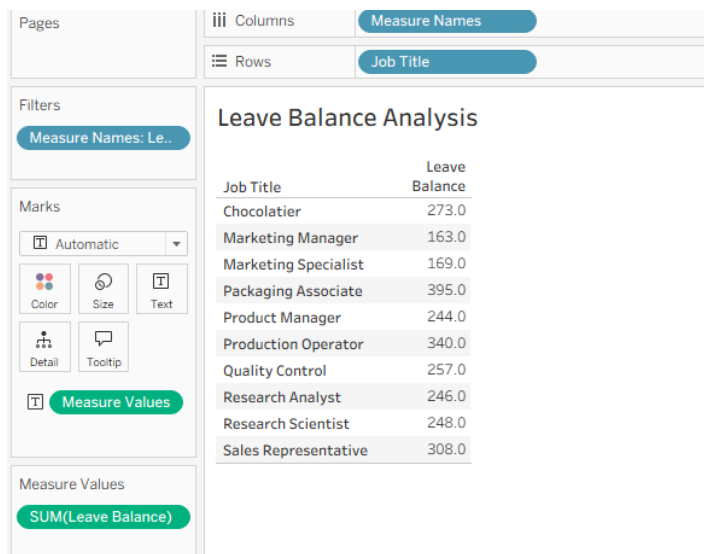
a) People in each job



b) Gender distribution

c)



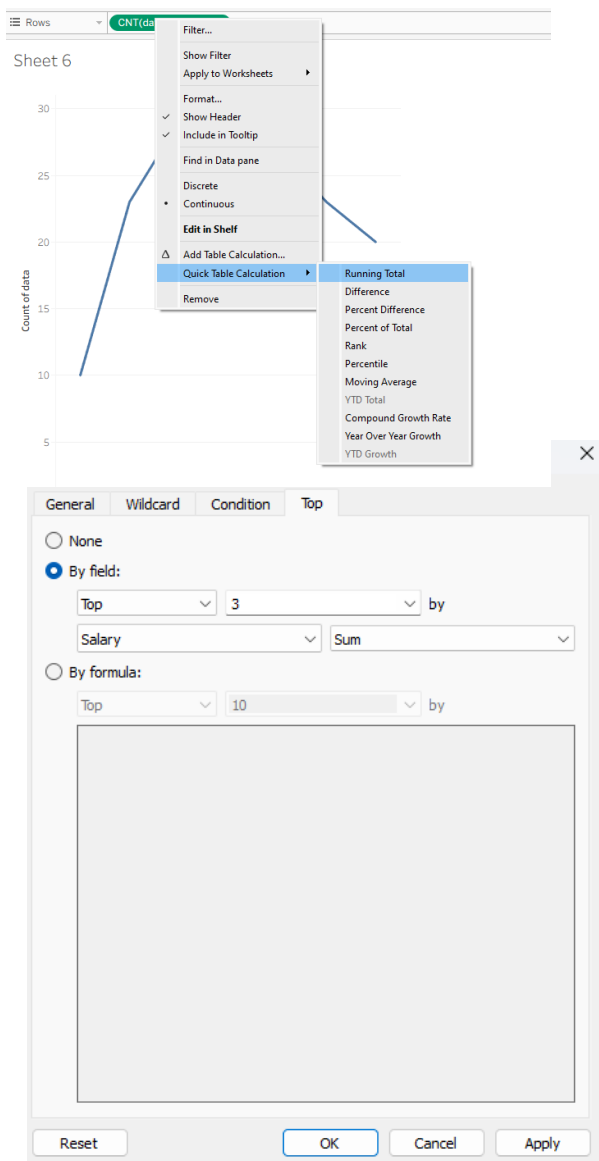


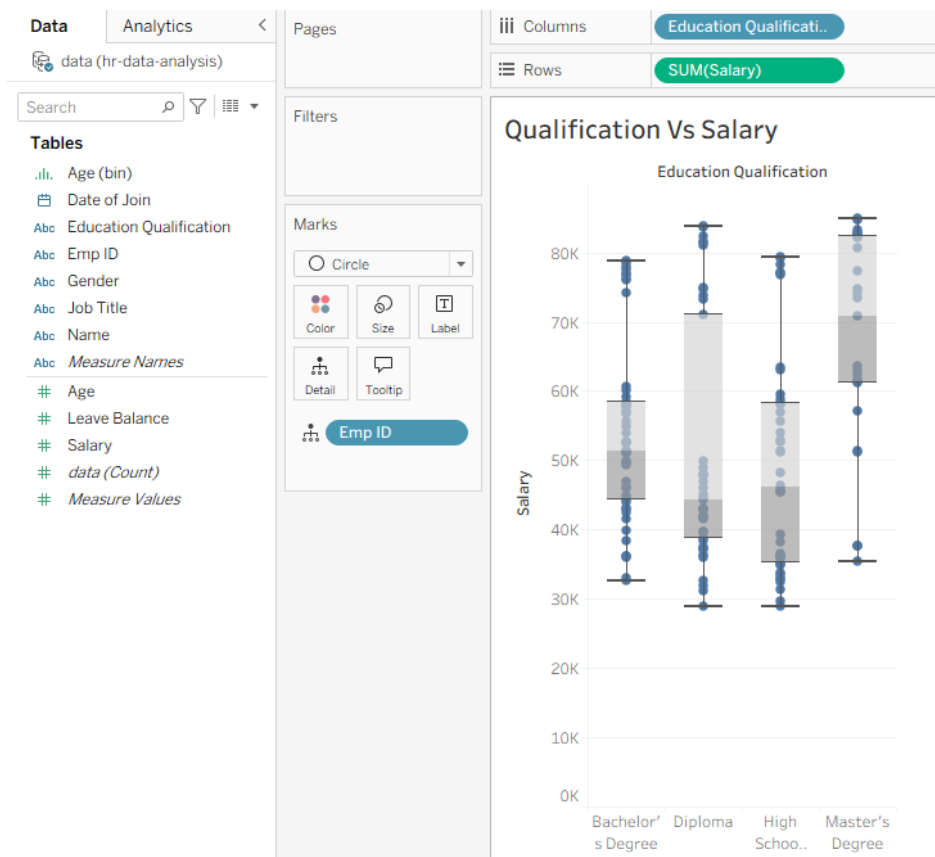
d)

e)

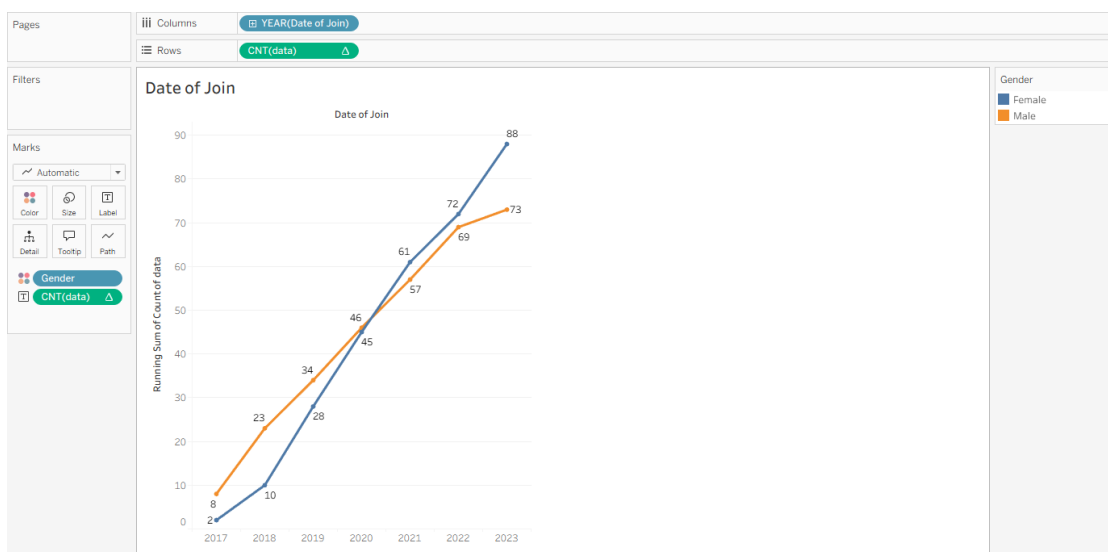


f)

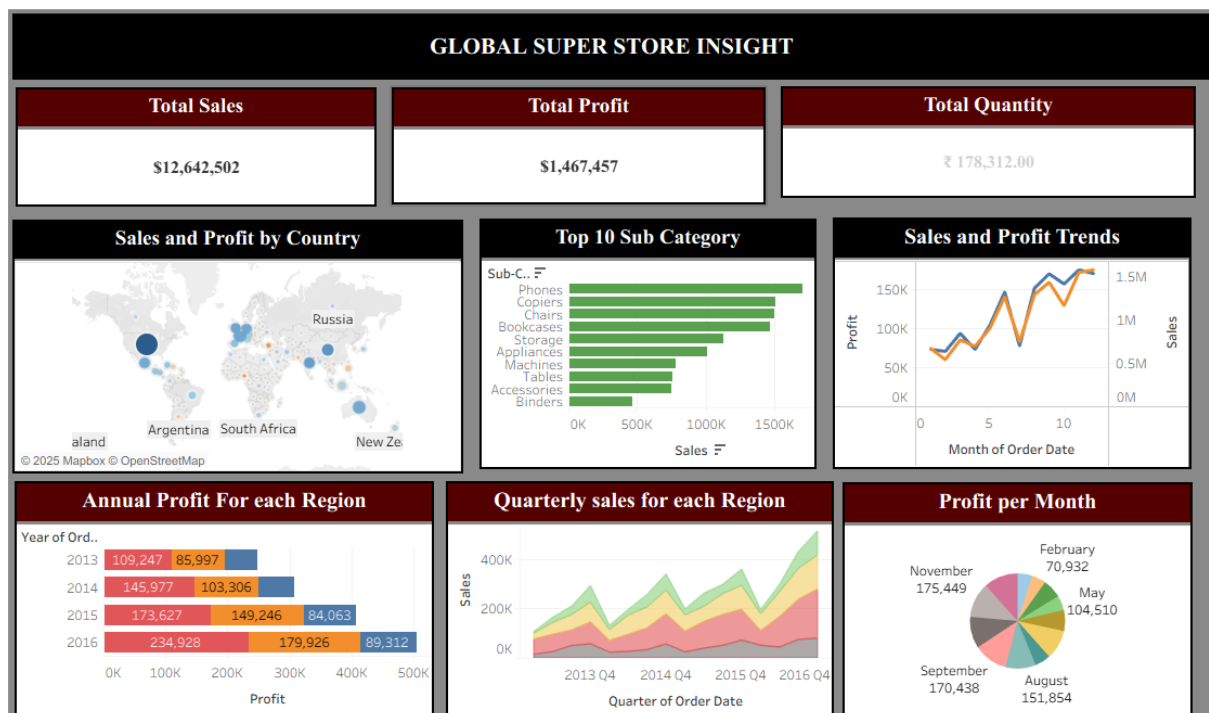
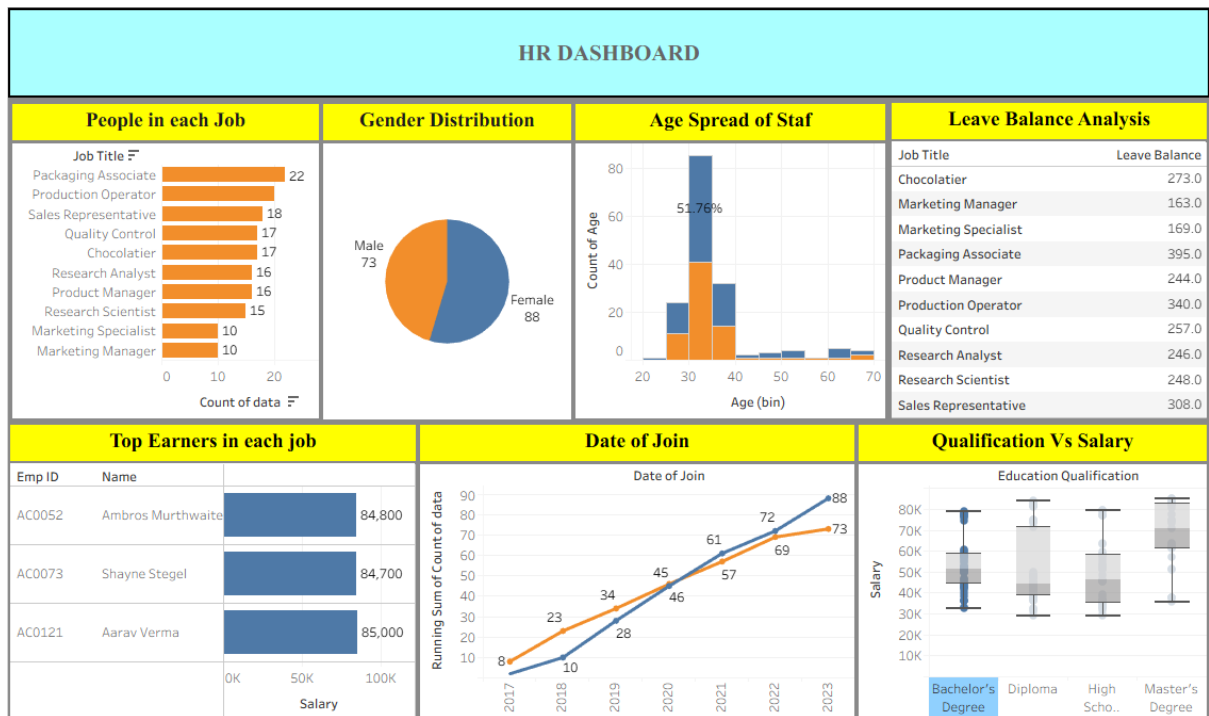




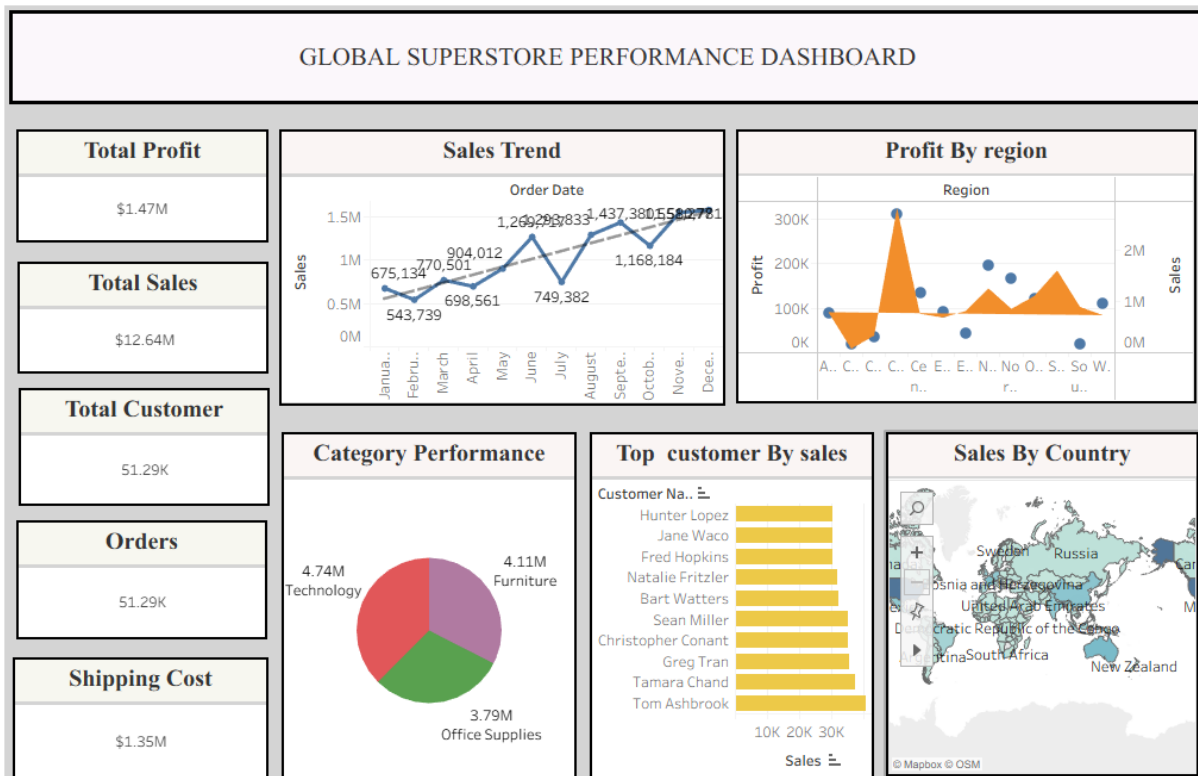
g)

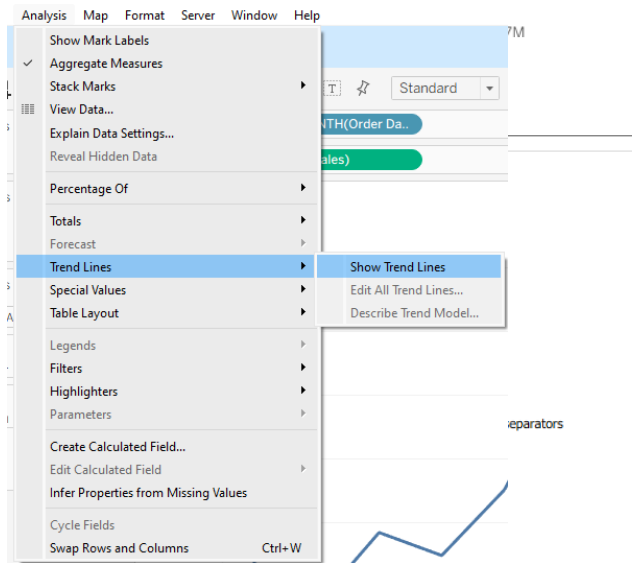


h)



P3)





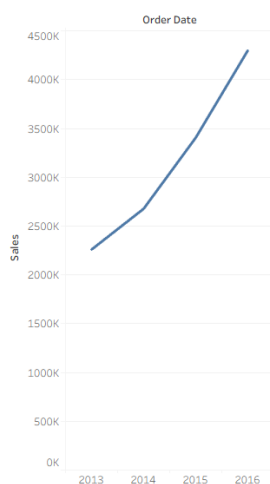
a)

Format SUM(Profit)		Pages	Columns	Rows
A	Fields			
Axis	Pane	Filters		
Default				
Font: Tahlequah Book				
			Total Profit	
			\$1.47M	

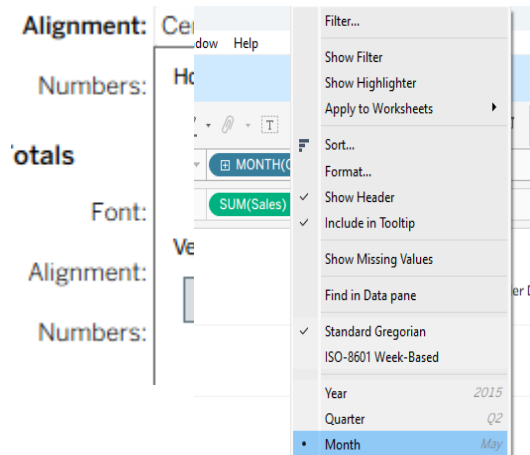
b)

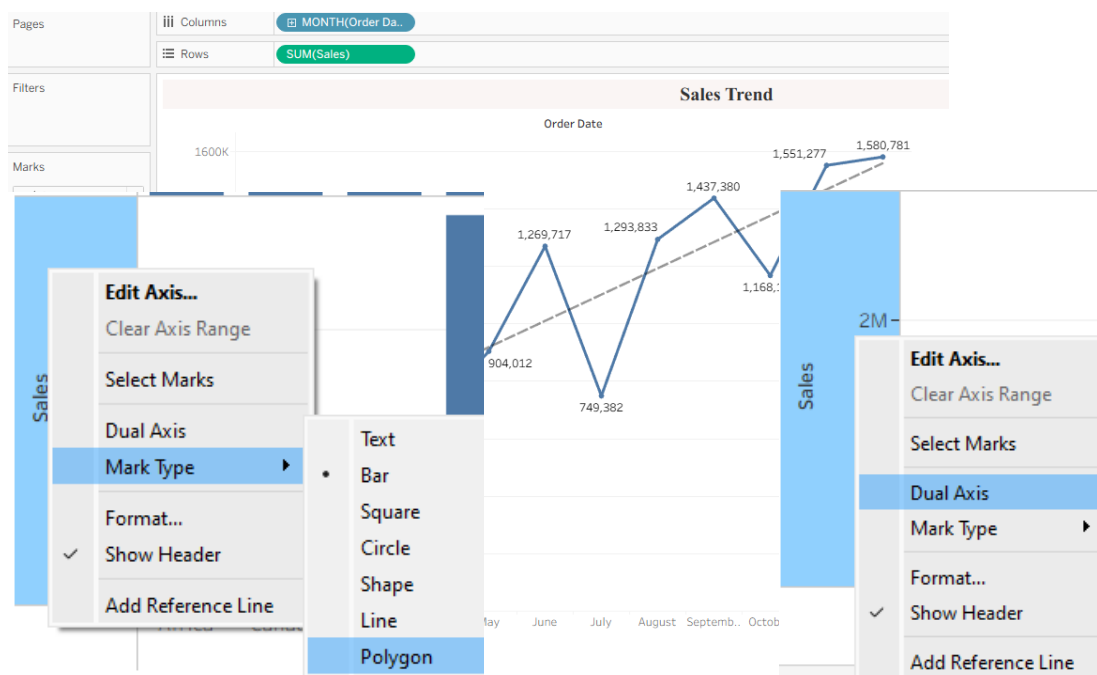
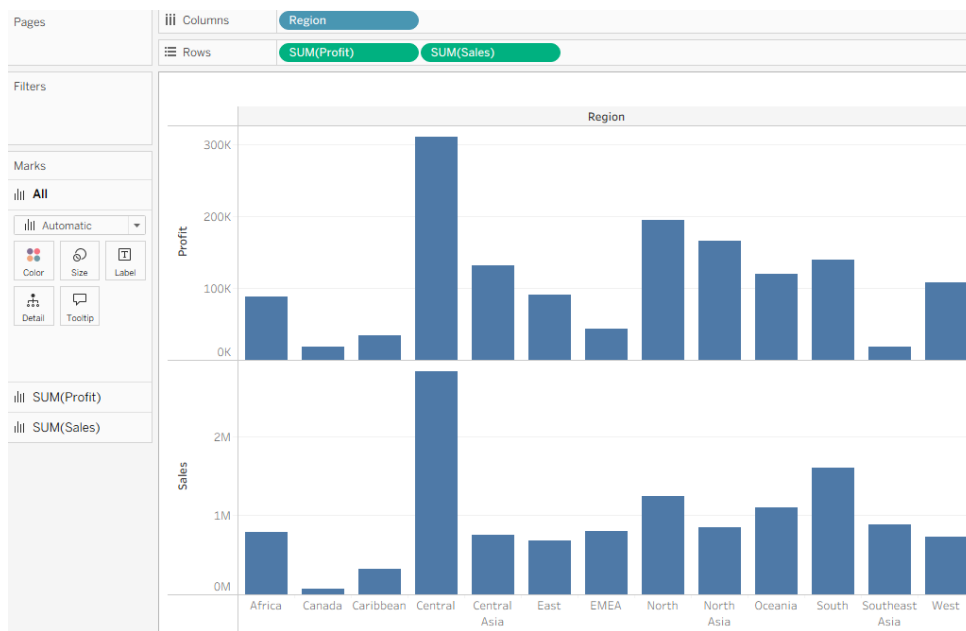
Format SUM(Profit)		Pages	Columns	Rows
A	Fields			
Axis	Pane	Filters		
Default				
Font:				
			Total Sales	
			\$12.64M	

Columns	YEAR(Order Date)
Rows	SUM(Sales)

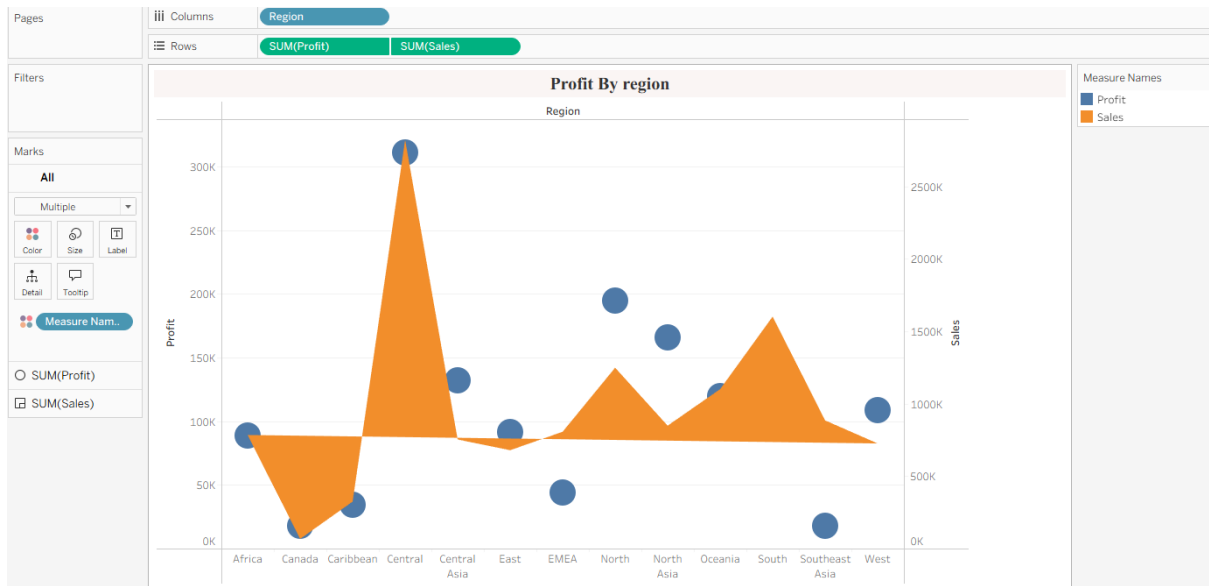


c)

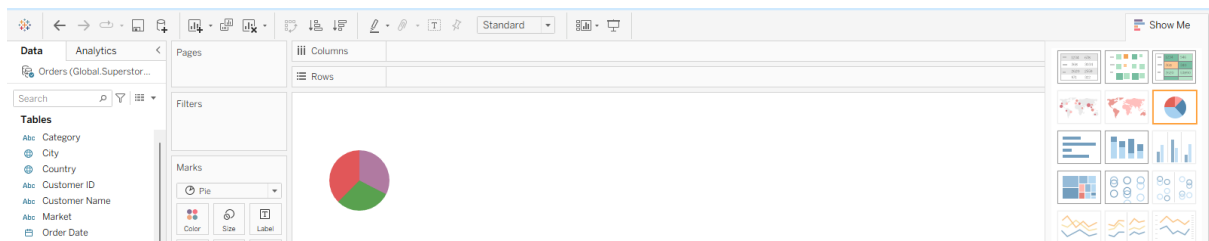
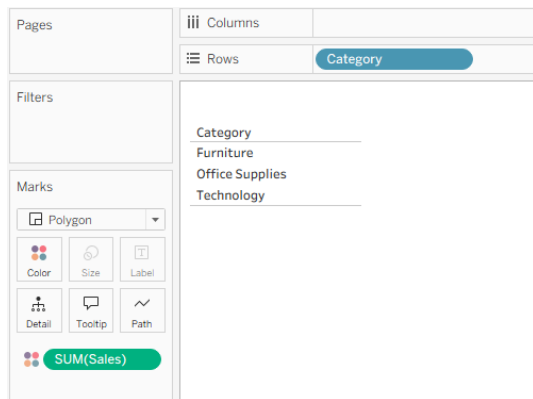


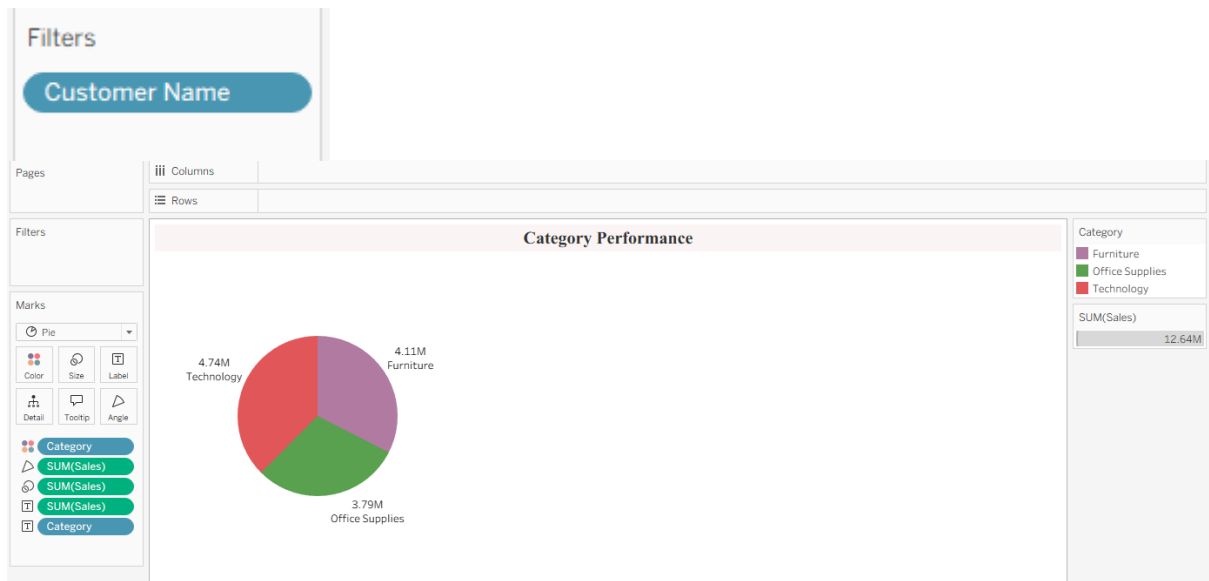


d)

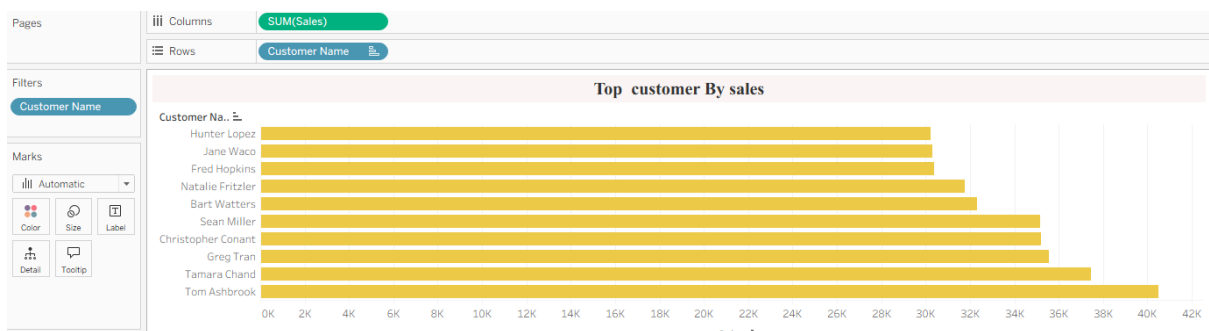


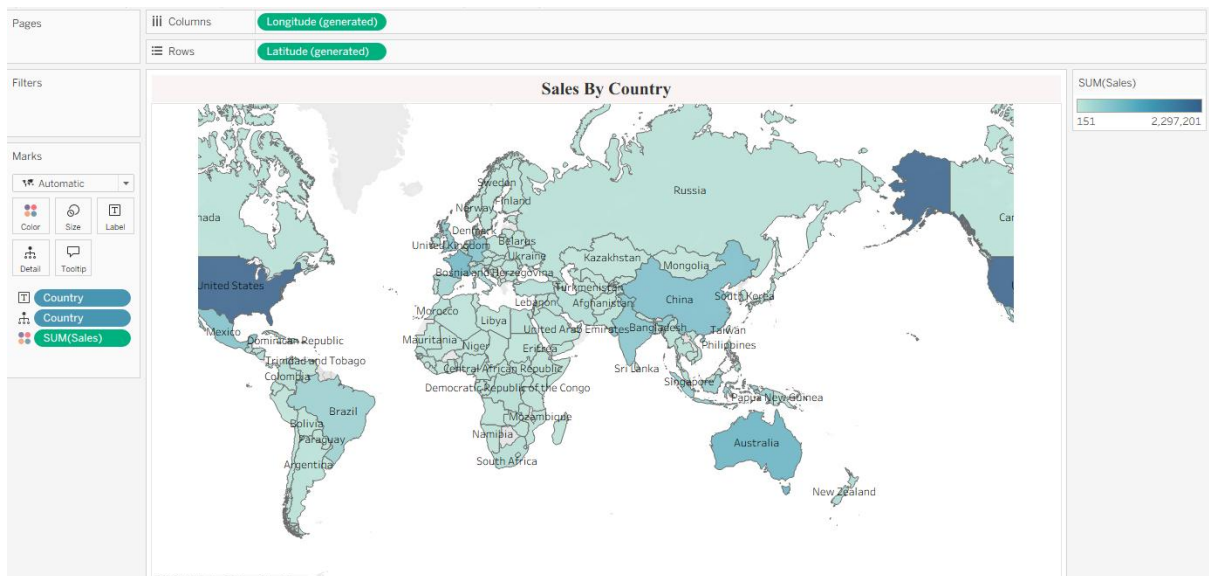
d)





e)





Filter [Customer Name]

General

Wildcard

Condition

Top

☐ None
 ☒ By field:

Top

10

by

Sales

Sum

☐ By formula:

Top

10

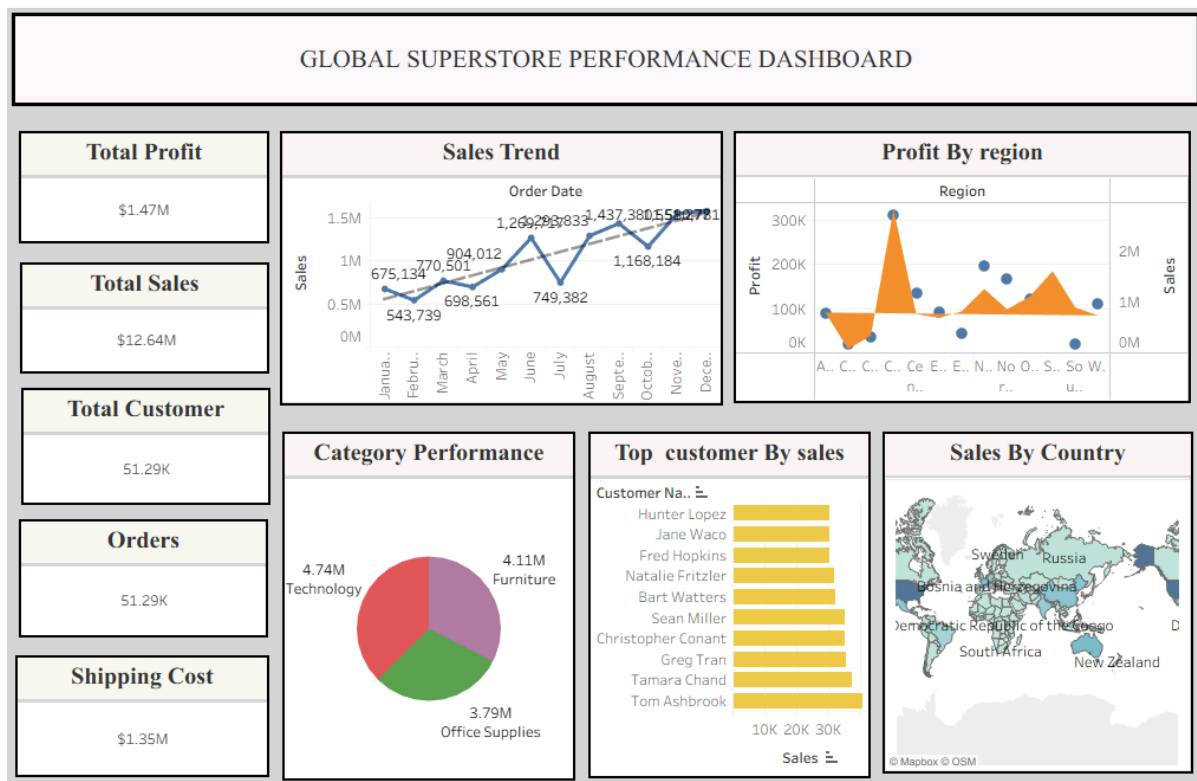
by

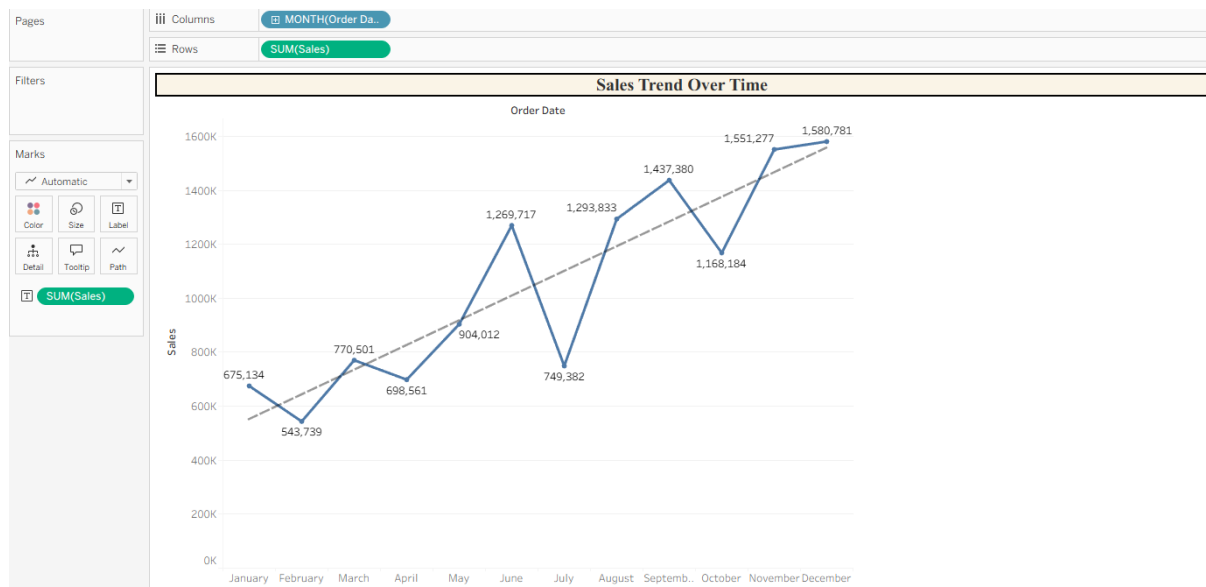
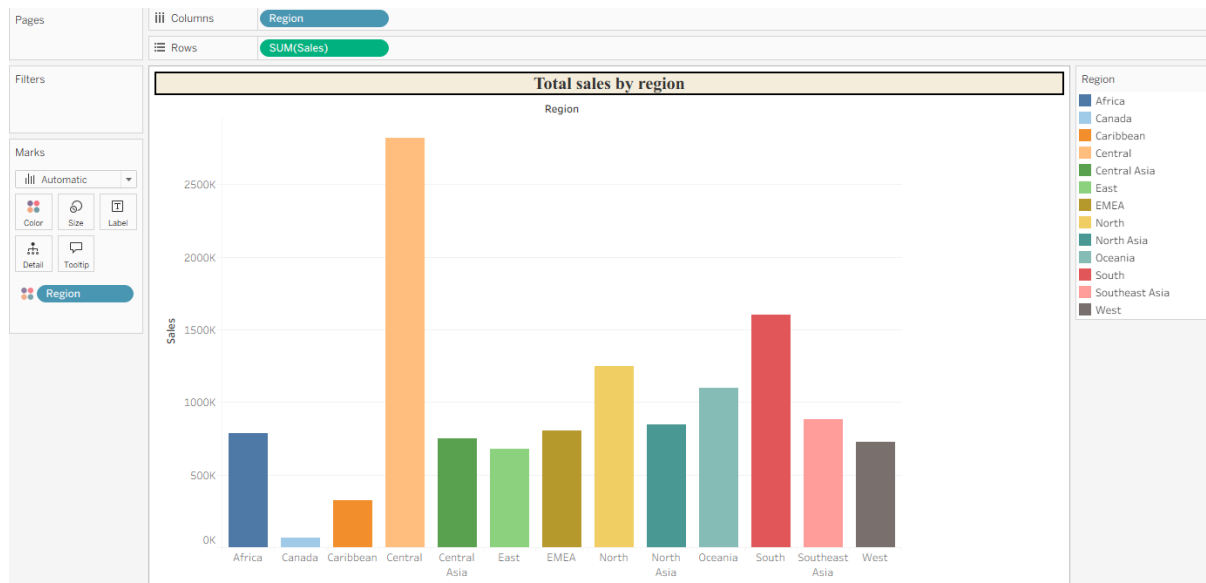
Reset

OK

Cancel

Apply





4.Create the given global superstore highlights Dashboard using global superstore data.

